

MTH251 & MTH252

# **Mathematical Analysis I & II**

Course Textbook

Department of Mathematical Sciences  
**UNIST**

## Mathematical Analysis I & II

Course Textbook

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### Source Code & Contributions

GitHub: <https://github.com/sihyeon8240/Math>

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# Contents

|          |                                                 |           |
|----------|-------------------------------------------------|-----------|
| <b>1</b> | <b>The Number Systems</b>                       | <b>1</b>  |
| 1.1      | The Natural Number Systems . . . . .            | 2         |
| 1.2      | Relations . . . . .                             | 4         |
| 1.3      | Ordered Sets . . . . .                          | 8         |
| 1.4      | Fields and Ordered Fields . . . . .             | 12        |
| 1.5      | The Real Field . . . . .                        | 13        |
| 1.6      | The Extended Real Number System . . . . .       | 21        |
| 1.7      | The Complex Field . . . . .                     | 23        |
| 1.8      | Euclidean Spaces . . . . .                      | 26        |
| <b>2</b> | <b>Basic Topology</b>                           | <b>28</b> |
| 2.1      | Functions . . . . .                             | 28        |
| 2.2      | Finite Sets . . . . .                           | 31        |
| 2.3      | Countable and Uncountable Sets . . . . .        | 40        |
| 2.4      | Metric Spaces . . . . .                         | 49        |
| 2.5      | Compact Sets . . . . .                          | 59        |
| 2.6      | Connected Sets . . . . .                        | 70        |
| <b>3</b> | <b>Sequences and Series</b>                     | <b>72</b> |
| 3.1      | Convergent Sequences . . . . .                  | 72        |
| 3.2      | Subsequences . . . . .                          | 75        |
| 3.3      | Cauchy Sequences . . . . .                      | 75        |
| 3.4      | Upper and Lower Limits . . . . .                | 75        |
| 3.5      | Some Special Sequences . . . . .                | 75        |
| 3.6      | Series . . . . .                                | 75        |
| 3.7      | Series of Nonnegative Terms . . . . .           | 75        |
| 3.8      | The Number $e$ . . . . .                        | 75        |
| 3.9      | The Root and Ratio Tests . . . . .              | 75        |
| 3.10     | Power Series . . . . .                          | 75        |
| 3.11     | Summation by Parts . . . . .                    | 75        |
| 3.12     | Absolute Convergence . . . . .                  | 75        |
| 3.13     | Addition and Multiplication of Series . . . . . | 75        |
| 3.14     | Rearrangements . . . . .                        | 75        |

|          |                                                           |           |
|----------|-----------------------------------------------------------|-----------|
| <b>4</b> | <b>Continuity</b>                                         | <b>76</b> |
| 4.1      | Limits of Functions . . . . .                             | 76        |
| 4.2      | Continuous Functions . . . . .                            | 76        |
| 4.3      | Continuity and Compactness . . . . .                      | 76        |
| 4.4      | Continuity and Connectedness . . . . .                    | 76        |
| 4.5      | Discontinuities . . . . .                                 | 76        |
| 4.6      | Monotonic Functions . . . . .                             | 76        |
| 4.7      | Infinite Limits and Limits at Infinity . . . . .          | 76        |
| <b>5</b> | <b>Differentiation</b>                                    | <b>77</b> |
| 5.1      | The Derivative of a Real Function . . . . .               | 77        |
| 5.2      | Mean Value Theorems . . . . .                             | 77        |
| 5.3      | The Continuity of Derivatives . . . . .                   | 77        |
| 5.4      | L'Hospital's Rule . . . . .                               | 77        |
| 5.5      | Derivatives of Higher Order . . . . .                     | 77        |
| 5.6      | Taylor's Theorem . . . . .                                | 77        |
| 5.7      | Differentiation of Vector-valued Functions . . . . .      | 77        |
| <b>6</b> | <b>The Riemann-Stieltjes Integral</b>                     | <b>78</b> |
| 6.1      | Definition and Existence of the Integral . . . . .        | 78        |
| 6.2      | Properties of the Integral . . . . .                      | 78        |
| 6.3      | Integration and Differentiation . . . . .                 | 78        |
| 6.4      | Integration of Vector-valued Functions . . . . .          | 78        |
| 6.5      | Rectifiable Curves . . . . .                              | 78        |
| <b>7</b> | <b>Sequences and Series of Functions</b>                  | <b>79</b> |
| 7.1      | Discussion of Main Problem . . . . .                      | 79        |
| 7.2      | Uniform Convergence . . . . .                             | 79        |
| 7.3      | Uniform Convergence and Continuity . . . . .              | 79        |
| 7.4      | Uniform Convergence and Integration . . . . .             | 79        |
| 7.5      | Uniform Convergence and Differentiation . . . . .         | 79        |
| 7.6      | Equicontinuous Families of Functions . . . . .            | 79        |
| 7.7      | Equicontinuous Families of Functions . . . . .            | 79        |
| 7.8      | The Stone-Weierstrass Theorem . . . . .                   | 79        |
| <b>8</b> | <b>Some Special Functions</b>                             | <b>80</b> |
| 8.1      | Power Series . . . . .                                    | 80        |
| 8.2      | The Exponential and Logarithmic Functions . . . . .       | 80        |
| 8.3      | The Trigonometric Functions . . . . .                     | 80        |
| 8.4      | The Algebraic Completeness of the Complex Field . . . . . | 80        |
| 8.5      | Fourier Series . . . . .                                  | 80        |
| 8.6      | The Gamma Function . . . . .                              | 80        |

|           |                                                          |           |
|-----------|----------------------------------------------------------|-----------|
| <b>9</b>  | <b>Functions of Several Variables</b>                    | <b>81</b> |
| 9.1       | Linear Transformations . . . . .                         | 81        |
| 9.2       | Differentiation . . . . .                                | 81        |
| 9.3       | The Contraction Principle . . . . .                      | 81        |
| 9.4       | The Inverse Function Theorem . . . . .                   | 81        |
| 9.5       | The Implicit Function Theorem . . . . .                  | 81        |
| 9.6       | The Rank Theorem . . . . .                               | 81        |
| 9.7       | Determinants . . . . .                                   | 81        |
| 9.8       | Derivatives of Higher Order . . . . .                    | 81        |
| 9.9       | Differentiation of Integrals . . . . .                   | 81        |
| <b>10</b> | <b>Integration of Differential Forms</b>                 | <b>82</b> |
| 10.1      | Integration . . . . .                                    | 82        |
| 10.2      | Primitive Mappings . . . . .                             | 82        |
| 10.3      | Partitions of Unity . . . . .                            | 82        |
| 10.4      | Change of Variables . . . . .                            | 82        |
| 10.5      | Differential Forms . . . . .                             | 82        |
| 10.6      | Simplexes and Chains . . . . .                           | 82        |
| 10.7      | Stokes' Theorem . . . . .                                | 82        |
| 10.8      | Closed Forms and Exact Forms . . . . .                   | 82        |
| 10.9      | Vector Analysis . . . . .                                | 82        |
| <b>11</b> | <b>The Lebesgue Theory</b>                               | <b>83</b> |
| 11.1      | Set Functions . . . . .                                  | 83        |
| 11.2      | Construction of the Lebesgue Measure . . . . .           | 83        |
| 11.3      | Measure Spaces . . . . .                                 | 83        |
| 11.4      | Measurable Functions . . . . .                           | 83        |
| 11.5      | Simple Functions . . . . .                               | 83        |
| 11.6      | Integration . . . . .                                    | 83        |
| 11.7      | Comparison with the Riemann Integral . . . . .           | 83        |
| 11.8      | Integration of Complex Functions . . . . .               | 83        |
| 11.9      | Functions of Class $\mathcal{L}^2$ . . . . .             | 83        |
|           | <b>Appendix: Construction of <math>\mathbb{R}</math></b> | <b>84</b> |

# Chapter 1

## The Number Systems

*Remark.* Throughout this textbook, **logical formulas** are interpreted according to the following conventions:

- **Parentheses** have the highest precedence. For example,

$$(P \vee Q) \wedge R$$

is evaluated by first computing  $P \vee Q$

- The **logical connectives** are interpreted in the order

$$\neg > \wedge > \vee > \implies > \iff$$

Hence

$$\neg P \wedge Q \vee R \implies S \iff T$$

is understood as

$$(((\neg P) \wedge Q) \vee R) \implies S) \iff T$$

- A **quantifier** binds the entire formula that follows it unless parentheses indicate a smaller scope. Thus

$$\forall x, P(x) \wedge Q(x) \implies R(x)$$

means

$$\forall x ((P(x) \wedge Q(x)) \implies R(x))$$

whereas

$$(\forall x P(x)) \wedge (\forall x Q(x))$$

is different from

$$\forall x (P(x) \wedge Q(x))$$

## 1.1 The Natural Number Systems

The set  $\{1, 2, 3, \dots\}$  of natural numbers is denoted by  $\mathbb{N}$ , and is defined by the following axioms:

- $1 \in \mathbb{N}$
- $\forall n \in \mathbb{N}, s(n) \in \mathbb{N}$
- $\nexists n \in \mathbb{N}, s(n) = 1$
- $\forall m, n \in \mathbb{N}, s(m) = s(n) \implies m = n$
- If  $S \subseteq \mathbb{N}$  satisfies
  - $1 \in S$
  - $\forall n \in S, s(n) \in S$

then  $S = \mathbb{N}$

where  $s(n)$  is called the **successor** of  $n$ . These axioms are called the **Peano axioms** (or *Peano Postulates*).

*Remark.* The **Peano axioms** do not guarantee the *existence* of  $\mathbb{N}$ , which is guaranteed by the **axiom of infinity** in the **ZFC axioms**. In this textbook, the **ZFC axioms** is not discussed, so we will just assume that  $\mathbb{N}$  exists.

**Definition 1.1.1.** Let  $P(n)$  be a statement about  $n \in \mathbb{N}$ , and let

$$S := \{n \in \mathbb{N} \mid P(n) \text{ is true}\} \subseteq \mathbb{N}$$

Assume that  $P(n)$  satisfies

- $P(1)$  is true
- $P(n)$  is true  $\implies P(n + 1)$  is true

This implies that

- $1 \in S$
- $n \in S \implies n + 1 \in S$

Therefore,

$$\begin{aligned} S &= \mathbb{N} \\ \Downarrow \\ P(n) &\text{ is true for all } n \in \mathbb{N} \end{aligned}$$

This method of proof is called the **principle of mathematical induction**.

**Definition 1.1.2.** For all  $n, m \in \mathbb{N}$ , we define the **addition** operator  $+$  on  $\mathbb{N}$  by

- $n + 1 := s(n)$
- $n + s(m) := s(n + m)$

For all  $n, m \in \mathbb{N}$ , we define the **multiplication** operator  $\cdot$  on  $\mathbb{N}$  by

- $n \cdot 1 := n$
- $n \cdot s(m) := n \cdot m + n$

*Remark.* Strictly speaking, [Definition 1.1.2](#) requires the **recursion theorem** to be strictly defined. In this textbook, the **recursion theorem** is not discussed, so we will just assume that the operators on  $\mathbb{N}$  are well-defined.

Also, we will not discuss the properties of the operators on  $\mathbb{N}$ , such as

- $n + m = m + n$
- $(n + m) + k = n + (m + k)$
- $n \cdot m = m \cdot n$
- $(n \cdot m) \cdot k = n \cdot (m \cdot k)$
- $n \cdot (m + k) = n \cdot m + n \cdot k$

for all  $n, m, k \in \mathbb{N}$ . For these proofs rely on the recursive definitions.

**Theorem 1.1.3.** Let  $S \subseteq \mathbb{N}$  be nonempty. Then  $S$  has a least element. This theorem is called the **well-ordering property** of  $\mathbb{N}$ .

*Proof.* Assume that  $S$  has no least element. Let

- $T := \mathbb{N} \setminus S$
- $L := \{n \in \mathbb{N} \mid \{1, 2, \dots, n\} \subseteq T\}$

We will show that

$$\begin{array}{c}
 L = \mathbb{N} \\
 \Downarrow \\
 T = \mathbb{N} \\
 \Downarrow \\
 S = \emptyset \\
 \Downarrow \\
 \perp
 \end{array}$$

- **Base case:** If  $1 \in S$ , then

$$\begin{aligned} 1 &\in S \\ \Downarrow \\ 1 &= \min S \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\begin{aligned} 1 &\notin S \\ \Downarrow \\ 1 &\in T \\ \Downarrow \\ 1 &\in L \end{aligned}$$

- **Inductive step:** Assume that  $n \in L$  for some  $n \in \mathbb{N}$ . Then

$$\begin{aligned} n &\in L \\ \Downarrow \\ \{1, 2, \dots, n\} &\subseteq T \\ \Downarrow \\ \{1, 2, \dots, n\} \cap S &= \emptyset \end{aligned}$$

If  $n + 1 \in S$ , then

$$\begin{aligned} n + 1 &\in S \\ \Downarrow \\ n + 1 &= \min S \\ \Downarrow \\ &\perp \end{aligned}$$

Therefore,

$$\begin{aligned} n + 1 &\notin S \\ \Downarrow \\ n + 1 &\in T \\ \Downarrow \\ \{1, 2, \dots, n + 1\} &\subseteq T \\ \Downarrow \\ n + 1 &\in L \end{aligned}$$

By the *principle of mathematical induction*,  $L = \mathbb{N}$ . □

## 1.2 Relations

Let  $S$  be a set. A **relation**  $\sim$  on  $S$  is a subset of  $S \times S$ . Equivalently, a subset  $R$  of  $S \times S$  is a relation on  $S$ .

- $R = \{(x, y) \in S \times S \mid x \sim y\}$

$$\bullet \ x \sim y \iff (x, y) \in R$$

**Definition 1.2.1.** Let  $S$  be a set. A relation  $\sim$  on  $S$  is called an **equivalence relation** if it satisfies the following properties:

- **Reflexivity:**  $\forall x \in S, x \sim x$
- **Symmetry:**  $\forall x, y \in S, x \sim y \implies y \sim x$
- **Transitivity:**  $\forall x, y, z \in S, (x \sim y \wedge y \sim z) \implies x \sim z$

**Definition 1.2.2.** Let  $S$  be a set. A collection  $\{A_\alpha\}_{\alpha \in I}$  of sets is called a **partition** of  $S$  if

- $\forall \alpha \in I, (A_\alpha \neq \emptyset \wedge A_\alpha \subseteq S)$
- $\forall \alpha, \beta \in I, (\alpha \neq \beta \implies A_\alpha \cap A_\beta = \emptyset)$
- $\bigcup_{\alpha \in I} A_\alpha = S$

**Definition 1.2.3.** Let  $S$  be a set, and let  $\sim$  be an equivalence relation on  $S$ . For each  $x \in S$ , the **equivalence class** of  $x$  is defined by

$$[x] := \{y \in S \mid y \sim x\}$$

We denote the set of all equivalence classes of  $S$  by

$$S/\sim := \{[x] \mid x \in S\}$$

**Theorem 1.2.4.** Let  $S$  be a set, and let  $\sim$  be an equivalence relation on  $S$ . Then

$$\forall x, y \in S, [x] \cap [y] \neq \emptyset \implies [x] = [y]$$

*Proof.* Assume that  $[x] \cap [y] \neq \emptyset$ . Then

$$\begin{aligned} \exists z \in S, z \in [x] \cap [y] \\ \Downarrow \\ z \in [x] \wedge z \in [y] \\ \Downarrow \\ z \sim x \wedge z \sim y \\ \Downarrow \\ x \sim z \wedge z \sim y \\ \Downarrow \\ x \sim y \end{aligned}$$

Therefore, for any  $w \in [x]$ ,

$$\begin{aligned} w &\sim x \wedge x \sim y \\ &\Downarrow \\ w &\sim y \\ &\Downarrow \\ w &\in [y] \end{aligned}$$

Therefore,

$$[x] \subseteq [y]$$

Similarly,

$$[y] \subseteq [x]$$

□

**Theorem 1.2.5.** Let  $S$  be a set, and let  $\{A_\alpha\}_{\alpha \in I}$  be a partition of  $S$ . Let

$$x \sim y \iff \exists \alpha \in I, (x \in A_\alpha \wedge y \in A_\alpha)$$

for all  $x, y \in S$ . Then

(A)  $\sim$  is an equivalence relation on  $S$

(B)  $\{A_\alpha\}_{\alpha \in I} = S / \sim$

*Proof.* (A) Let  $x, y, z \in S$ .

• **Reflexivity:** For any  $x$ ,

$$\begin{aligned} x &\in S \\ &\Downarrow \\ x &\in \bigcup_{\alpha \in I} A_\alpha \\ &\Downarrow \\ \exists \alpha \in I, x &\in A_\alpha \\ &\Downarrow \\ x &\sim x \end{aligned}$$

• **Symmetry:** If  $x \sim y$ , then

$$\begin{aligned} \exists \alpha \in I, (x &\in A_\alpha \wedge y \in A_\alpha) \\ &\Downarrow \\ y &\in A_\alpha \wedge x \in A_\alpha \\ &\Downarrow \\ y &\sim x \end{aligned}$$

• **Transitivity:** If  $x \sim y \wedge y \sim z$ , then

$$\begin{aligned}
 & \exists \alpha \in I, (x \in A_\alpha \wedge y \in A_\alpha) \\
 & \exists \beta \in I, (y \in A_\beta \wedge z \in A_\beta) \\
 & \Downarrow \\
 & y \in A_\alpha \wedge y \in A_\beta \\
 & \Downarrow \\
 & A_\alpha \cap A_\beta \neq \emptyset \\
 & \Downarrow \\
 & \alpha = \beta \\
 & \Downarrow \\
 & x \in A_\alpha \wedge z \in A_\alpha \\
 & \Downarrow \\
 & x \sim z
 \end{aligned}$$

(B) Let  $x \in S$ . Then

$$\begin{aligned}
 & x \in S \\
 & \Downarrow \\
 & x \in \bigcup_{\alpha \in I} A_\alpha \\
 & \Downarrow \\
 & \exists \alpha \in I, x \in A_\alpha \\
 & \Downarrow \\
 & x \in [x] \\
 & \Downarrow \\
 & \exists \alpha \in I, A_\alpha = [x]
 \end{aligned}$$

Therefore,

$$\{A_\alpha\}_{\alpha \in I} \subseteq S / \sim$$

Let  $[x] \in S / \sim$ . Then

$$\begin{aligned}
 & x \in S \\
 & \Downarrow \\
 & x \in \bigcup_{\alpha \in I} A_\alpha \\
 & \Downarrow \\
 & \exists \alpha \in I, x \in A_\alpha \\
 & \Downarrow \\
 & A_\alpha \cap [x] \neq \emptyset \\
 & \Downarrow \\
 & A_\alpha = [x]
 \end{aligned}$$

Therefore,

$$S / \sim \subseteq \{A_\alpha\}_{\alpha \in I}$$

□

### 1.3 Ordered Sets

A set  $S$  is called an **ordered set** if  $S$  is equipped with a relation  $<$  such that for any  $x, y, z \in S$ , the following properties hold:

$$(A) \quad x < y, y < z \implies x < z$$

$$(B) \quad \text{Only one of the relations } x < y, x = y, \text{ or } x > y \text{ holds}$$

the relation  $<$  is called a **(strict) total order** on  $S$ .

On an ordered set, we define the following relations:

- $x > y \iff y < x$
- $x \geq y \iff (x > y \text{ or } x = y)$
- $x \leq y \iff y \geq x$

**Example 1.3.1.**  $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$  are ordered sets with the following relations:

- $\forall n, m \in \mathbb{N}, m < n \iff \exists k \in \mathbb{N} \text{ such that } n = m + k$
- $\forall n, m \in \mathbb{Z}, m < n \iff \exists k \in \mathbb{Z}^+ \text{ such that } n = m + k$
- $\forall p, q \in \mathbb{Q}, q < p \iff \exists r \in \mathbb{Q}^+ \text{ such that } p = q + r$

**Definition 1.3.2.** Let  $S$  be an ordered set and  $E \subseteq S$ .

- $E$  is said to be **bounded above** if

$$\exists M \in S \text{ such that } \forall x \in E, x \leq M$$

Such an  $M$  is called an **upper bound** of  $E$ .

- $E$  is said to be **bounded below** if

$$\exists m \in S \text{ such that } \forall x \in E, m \leq x$$

Such an  $m$  is called a **lower bound** of  $E$ .

**Theorem 1.3.3.** Let  $E \subseteq \mathbb{Z}$  be nonempty.

(A)  $E$  is bounded below  $\implies E$  has a least element.

(B)  $E$  is bounded above  $\implies E$  has a greatest element.

*Proof.* (A) Let  $E$  be bounded below. Then,

$$\begin{aligned} &E \text{ is bounded below} \\ &\Downarrow \\ &\exists m \in \mathbb{Z} \text{ such that } \forall x \in E, m \leq x \end{aligned}$$

Let

$$F := \{x - m + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{aligned} &E \neq \emptyset \\ &\Downarrow \\ &F \neq \emptyset \end{aligned}$$

By [Theorem 1.1.3](#),  $F$  has a least element, say  $n$ . Then

$$\begin{aligned} &\forall x \in E, n \leq x - m + 1 \\ &\Downarrow \\ &\forall x \in E, n + m - 1 \leq x \end{aligned}$$

Since  $n = \min F$ ,

$$\begin{aligned} &n \in F \\ &\Downarrow \\ &\exists x_0 \in E \text{ such that } n = x_0 - m + 1 \\ &\Downarrow \\ &x_0 = n + m - 1 = \min E \end{aligned}$$

(B) Let  $E$  be bounded above. Then,

$$\begin{aligned} &E \text{ is bounded above} \\ &\Downarrow \\ &\exists M \in \mathbb{Z} \text{ such that } \forall x \in E, x \leq M \end{aligned}$$

Let

$$F := \{-x + M + 1 \mid x \in E\} \subseteq \mathbb{N}$$

Then

$$\begin{aligned} &E \neq \emptyset \\ &\Downarrow \\ &F \neq \emptyset \end{aligned}$$

By [Theorem 1.1.3](#),  $F$  has a least element, say  $n$ . Then

$$\begin{aligned} &\forall x \in E, n \leq -x + M + 1 \\ &\Downarrow \\ &\forall x \in E, x \leq M - n + 1 \end{aligned}$$

Since  $n = \min F$ ,

$$\begin{aligned} n &\in F \\ \Downarrow \\ \exists x_0 \in E \text{ such that } n &= -x_0 + M + 1 \\ \Downarrow \\ x_0 &= M - n + 1 = \max E \end{aligned}$$

□

**Definition 1.3.4.** Let  $S$  be an ordered set, and let  $E \subseteq S$  be bounded above. An element  $\alpha \in S$  is called the **supremum** (or *least upper bound*) of  $E$  if

- $\alpha$  is an upper bound of  $E$
- $\forall \beta \in S, \beta < \alpha \implies \beta$  is not an upper bound of  $E$

The supremum of  $E$  is denoted by

$$\alpha = \sup E$$

Similarly, let  $E \subseteq S$  be bounded below. An element  $\alpha \in S$  is called the **infimum** (or *greatest lower bound*) of  $E$  if

- $\alpha$  is a lower bound of  $E$
- $\forall \beta \in S, \beta > \alpha \implies \beta$  is not a lower bound of  $E$

The infimum of  $E$  is denoted by

$$\alpha = \inf E$$

**Example 1.3.5.** Let

$$S := \{p \in \mathbb{Q}^+ \mid p^2 < 2\}$$

Let  $p \in S$  be given, and let

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} > p$$

Then

$$\begin{aligned} q^2 - 2 &= \frac{2(p^2 - 2)}{(p + 2)^2} < 0 \\ \Downarrow \\ q &\in S \end{aligned}$$

This implies that  $p \in S$  is not an upper bound of  $S$ .

Let  $p \in \mathbb{Q}^+ \setminus S$  be given. Then

$$p^2 \geq 2$$

$$\Downarrow \quad (\because \nexists p \in \mathbb{Q}^+, p^2 = 2)$$

$$p^2 > 2$$

$$\Downarrow \quad (\because \forall x \in S, x^2 < 2 < p^2 \implies x < p)$$

$p$  is an upper bound of  $S$

Let

$$q := p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2} < p$$

Then

$$q^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2} > 0$$

$$\Downarrow \quad (\because \forall x \in S, x^2 < 2 < q^2 \implies x < q)$$

$q$  is an upper bound of  $S$

This implies that  $p$  is not the least upper bound of  $S$ . To sum up,  $S$  has no least upper bound in  $\mathbb{Q}^+$ .

**Definition 1.3.6.** An ordered set  $S$  is said to have the **least upper bound property** if

$$\begin{aligned} &\forall E \subseteq S, E \text{ is nonempty and bounded above} \\ &\implies E \text{ has a least upper bound in } S. \end{aligned}$$

**Theorem 1.3.7.** Let  $S$  be an ordered set with the least upper bound property. Then

$$\begin{aligned} &\forall E \subseteq S, E \text{ is nonempty and bounded below} \\ &\implies E \text{ has a greatest lower bound in } S. \end{aligned}$$

*Proof.* Let  $E$  be a nonempty subset of  $S$  which is bounded below. Let  $L$  be defined as follows:

$$L := \{\lambda \mid \lambda \text{ is a lower bound of } E\}.$$

Then,  $L$  is nonempty and bounded above. By the least upper bound property, there exists  $\alpha = \sup L \in S$ . We will show that  $\alpha = \inf E$ .

Assume that  $\alpha \notin L$ . Then we have the following:

- $\alpha$  is not a lower bound of  $E \implies$  There exists  $\beta \in E$  such that  $\beta < \alpha$ .
- $\alpha = \sup L \implies \beta$  is not an upper bound of  $L \implies$  There exists  $\gamma \in L$  such that  $\beta < \gamma \leq \alpha$ .
- $\gamma \in L \implies \gamma$  is a lower bound of  $E \implies \gamma \leq \beta$ .

This is a contradiction. Therefore, we have  $\alpha \in L$ , i.e.,  $\alpha$  is a lower bound of  $E$ . Since  $\alpha$  is an upper bound of  $L$ , the following holds:

For any  $\delta > \alpha$ , we have  $\delta \notin L \implies \delta$  is not a lower bound of  $E$ .

Therefore, we can conclude that  $\alpha = \inf E$ . □

## 1.4 Fields and Ordered Fields

A set  $F$  is called a **field** if it is equipped with two operations, *addition* and *multiplication*, and satisfies the following axioms:

(A) Axioms for addition

- $x + y \in F$  for all  $x, y \in F$ .
- $x + y = y + x$  for all  $x, y \in F$ .
- $(x + y) + z = x + (y + z)$  for all  $x, y, z \in F$ .
- There exists an element  $0 \in F$  such that  $x + 0 = x$  for all  $x \in F$ .
- For each  $x \in F$ , there exists an element  $-x \in F$  such that  $x + (-x) = 0$ .

(B) Axioms for multiplication

- $x \cdot y \in F$  for all  $x, y \in F$ .
- $x \cdot y = y \cdot x$  for all  $x, y \in F$ .
- $(x \cdot y) \cdot z = x \cdot (y \cdot z)$  for all  $x, y, z \in F$ .
- There exists an element  $1 \in F$  with  $1 \neq 0$  such that  $x \cdot 1 = x$  for all  $x \in F$ .
- For each  $x \in F$  with  $x \neq 0$ , there exists an element  $x^{-1} \in F$  such that  $x \cdot x^{-1} = 1$ .

(C) The distributive law

- $x \cdot (y + z) = x \cdot y + x \cdot z$  for all  $x, y, z \in F$ .

A set  $F$  is called an **ordered field** if it is both an ordered set and a field, and satisfies the following axioms:

- If  $x, y, z \in F$  and  $x < y$ , then  $x + z < y + z$ .
- If  $x, y \in F$  and  $x > 0$  and  $y > 0$ , then  $x \cdot y > 0$ .

**Example 1.4.1.**  $(\mathbb{Q}, +, \cdot)$  is an ordered field, while  $(\mathbb{Z}, +, \cdot)$  is not an ordered field since  $2^{-1} \notin \mathbb{Z}$ .

## 1.5 The Real Field

There exists an ordered field  $(\mathbb{R}, +, \cdot)$  which has the *least upper bound property*. Moreover,  $\mathbb{R}$  contains  $\mathbb{Q}$  as a subfield, i.e.,  $\mathbb{Q} \subseteq \mathbb{R}$  and the operations of addition and multiplication on  $\mathbb{R}$  coincide with those on  $\mathbb{Q}$ .

We can construct  $\mathbb{R}$  from  $\mathbb{Q}$ . The construction of  $\mathbb{R}$  is rather complicated, so it will be discussed separately in [Appendix](#). In this section, we will just assume that  $\mathbb{R}$  exists.

**Theorem 1.5.1.** *For any  $x \in \mathbb{R}^+$  and  $y \in \mathbb{R}$ , there exists some  $n \in \mathbb{N}$  such that  $nx > y$ . This theorem is called the **Archimedean property** of  $\mathbb{R}$ .*

*Proof.* Let  $x \in \mathbb{R}^+$  and  $y \in \mathbb{R}$  be given. Let  $E$  be defined as follows:

$$E := \{nx \mid nx \leq y, n \in \mathbb{N}\}.$$

If  $E = \emptyset$ , which implies that  $x > y$ , then  $n = 1$  satisfies  $nx > y$ . Otherwise,  $E$  is nonempty and bounded above by  $y$ . By the least upper bound property, there exists  $\alpha = \sup E \in \mathbb{R}$ . Then we have the following:

- $\alpha - x$  is not an upper bound of  $E \implies$  There exists some  $m \in \mathbb{N}$  such that  $mx > \alpha - x$ .
- $mx > \alpha - x \implies (m+1)x > \alpha \implies (m+1)x \notin E \implies (m+1)x > y$ .

Therefore, we can take  $n = m + 1$  to complete the proof.  $\square$

**Corollary 1.5.2.** *For any  $\varepsilon \in \mathbb{R}^+$  and  $b \in (1, +\infty)$ , there exists some  $n \in \mathbb{N}$  such that  $b^{-n} < \varepsilon$ .*

*Proof.* For any  $n \in \mathbb{N}$  and  $x \in [-1, +\infty)$ , we have

$$(1+x)^n \geq 1+nx.$$

Let  $a \in \mathbb{R}^+$  and  $\varepsilon \in \mathbb{R}^+$  be given. By [Theorem 1.5.1](#), there exists some  $n \in \mathbb{N}$  such that  $na > \frac{1}{\varepsilon}$ . Then we have

$$(1+a)^n \geq 1+na > \frac{1}{\varepsilon} \implies (1+a)^{-n} < \varepsilon.$$

Since  $b > 1$ , we can take  $a = b - 1 > 0$  to complete the proof.  $\square$

**Theorem 1.5.3.** *Any nonempty subset of  $\mathbb{Z}$  which is bounded below in  $\mathbb{R}$  has a unique least element, and any nonempty subset of  $\mathbb{Z}$  which is bounded above in  $\mathbb{R}$  has a unique greatest element.*

*Proof.* Let  $E$  be a nonempty subset of  $\mathbb{Z}$  which is bounded below in  $\mathbb{R}$ . Then there exists a lower bound  $\alpha \in \mathbb{R}$  of  $E$ . Then we have the following:

- If  $\alpha \geq 0$ , then  $-1 \in \mathbb{Z}$  is a lower bound of  $E$ .
- If  $\alpha < 0$ , then by [Theorem 1.5.1](#), there exists some  $n \in \mathbb{N}$  such that  $n > -\alpha$ , which implies that  $-n \in \mathbb{Z}$  is a lower bound of  $E$ .

Therefore,  $E$  is bounded below in  $\mathbb{Z}$ . By [Theorem 1.3.3](#),  $E$  has a unique least element.

Let  $F$  be a nonempty subset of  $\mathbb{Z}$  which is bounded above in  $\mathbb{R}$ . Then there exists an upper bound  $\beta \in \mathbb{R}$  of  $F$ . Then we have the following:

- By [Theorem 1.5.1](#), there exists some  $n \in \mathbb{N}$  such that  $n > \beta$ , which implies that  $n \in \mathbb{Z}$  is an upper bound of  $F$ .

Therefore,  $F$  is bounded above in  $\mathbb{Z}$ . By [Theorem 1.3.3](#),  $F$  has a unique greatest element.  $\square$

**Definition 1.5.4.** For any  $x \in \mathbb{R}$ , the **floor function**  $\lfloor x \rfloor$  and **ceiling function**  $\lceil x \rceil$  are defined as follows:

- $\lfloor x \rfloor := \max\{k \in \mathbb{Z} \mid k \leq x\}$ .
- $\lceil x \rceil := \min\{k \in \mathbb{Z} \mid k \geq x\}$ .

[Theorem 1.5.3](#) guarantees the *existence* and *uniqueness* of  $\lfloor x \rfloor$  and  $\lceil x \rceil$ .

**Theorem 1.5.5.** *For any  $-\infty < x < y < +\infty$ , there exists some  $p \in \mathbb{Q}$  such that  $x < p < y$ . This theorem states that  $\mathbb{Q}$  is **dense** in  $\mathbb{R}$ .*

*Proof.* Let  $\varepsilon := y - x > 0$ . By [Theorem 1.5.1](#), there exists some  $n \in \mathbb{N}$  such that  $\frac{1}{n} < \varepsilon$ . Let  $S$  be defined as follows:

$$S := \left\{ k \in \mathbb{Z} \mid \frac{k}{n} < x \right\}.$$

$S$  is nonempty since  $n\lfloor x \rfloor - 1 \in S$ , and is bounded above by  $nx \in \mathbb{R}$ . By [Theorem 1.5.3](#),  $S$  has a greatest element, say  $m$ .

$$x < \frac{m+1}{n} < x + \frac{1}{n} < x + (y - x) = y.$$

Therefore, we can take  $p = \frac{m+1}{n} \in \mathbb{Q}$  to complete the proof.  $\square$

**Theorem 1.5.6.** *For each  $x \in \mathbb{R}^+$  and  $n \in \mathbb{N}$ , there exists a unique  $y \in \mathbb{R}^+$  such that  $y^n = x$ . We denote  $y$  by  $\sqrt[n]{x}$  or  $x^{\frac{1}{n}}$ .*

*Proof.* Let  $E$  be defined as follows:

$$E := \{t \in \mathbb{R}^+ \mid t^n < x\}.$$

$E$  is nonempty since  $2^{-1} \in E$  if  $x \geq 1$ , and  $x^2 \in E$  otherwise. Also,  $E$  is bounded above by  $\max\{x, 1\}$ . By the least upper bound property, there exists  $\alpha = \sup E \in \mathbb{R}^+$ . We will show that  $\alpha^n = x$ .

For any  $a, b \in \mathbb{R}^+$  with  $a < b$ , we have

$$b^n - a^n = (b - a) \sum_{k=0}^{n-1} b^{n-1-k} a^k < (b - a) n b^{n-1}.$$

Assume that  $\alpha^n < x$ . By [Theorem 1.5.5](#), there exists some  $h \in \mathbb{R}^+$  such that

$$h < \min\left\{1, \frac{x - \alpha^n}{n(\alpha + 1)^{n-1}}\right\}.$$

Put  $a = \alpha$  and  $b = \alpha + h$ . Then we have

$$\begin{aligned} (\alpha + h)^n - \alpha^n &< hn(\alpha + h)^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha + h)^n &< \alpha^n + hn(\alpha + 1)^{n-1} < \alpha^n + (x - \alpha^n) = x. \end{aligned}$$

Therefore, we have  $\alpha + h \in E$ , which is a contradiction.

Assume that  $\alpha^n > x$ . By [Theorem 1.5.5](#), there exists some  $h \in \mathbb{R}^+$  such that

$$h < \min\left\{1, \alpha, \frac{\alpha^n - x}{n(\alpha + 1)^{n-1}}\right\}.$$

Put  $a = \alpha - h$  and  $b = \alpha$ . Then we have

$$\begin{aligned} \alpha^n - (\alpha - h)^n &< hn\alpha^{n-1} < hn(\alpha + 1)^{n-1} \\ \implies (\alpha - h)^n &> \alpha^n - hn(\alpha + 1)^{n-1} > \alpha^n - (\alpha^n - x) = x. \end{aligned}$$

Therefore, we have  $\alpha - h$  is an upper bound of  $E$ , which is a contradiction.

Therefore, we have  $\alpha^n = x$ , which guarantees the *existence* of  $y$ . For any  $\alpha, \beta \in \mathbb{R}^+$ , we have  $\alpha \neq \beta \implies \alpha^n \neq \beta^n$ , which guarantees the *uniqueness* of  $y$ .  $\square$

**Definition 1.5.7.** Let  $y \in [0, 1)$  and  $b \in \mathbb{N}$  be given with  $b \geq 2$ . A sequence  $(a_n)_{n=1}^{\infty}$  is called the **greedy base  $b$  expansion** of  $y$  if for each  $n \in \mathbb{N}$ , we have

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n}$$

where  $a_k \in S_b := \{0, 1, 2, \dots, b-1\}$  for all  $k \in \mathbb{N}$ . Especially, if  $b = 10$ , the sequence is called the **greedy decimal expansion** of  $y$ .

**Theorem 1.5.8.** For any  $y \in [0, 1)$ , the greedy base  $b$  expansion of  $y$  exists and is unique.

*Proof.* Let  $(t_n)_{n=1}^{\infty}$ ,  $(T_n)_{n=1}^{\infty}$  and  $(a_n)_{n=1}^{\infty}$  be defined as follows:

- $t_1 := by$ ,  $T_1 := \{t \in S_b \mid t \leq t_1\}$ ,  $a_1 := \max T_1$ .
  - $t_2 := b^2y - ba_1$ ,  $T_2 := \{t \in S_b \mid t \leq t_2\}$ ,  $a_2 := \max T_2$ .
  - $\vdots$
  - $t_n := b^n y - \sum_{k=1}^{n-1} a_k b^{n-k}$ ,  $T_n := \{t \in S_b \mid t \leq t_n\}$ ,  $a_n := \max T_n$
- for all  $n \in \mathbb{N}$  with  $n \geq 2$ ,

Assume that  $0 \leq t_n < b$  for some  $n \in \mathbb{N}$ . Then, since  $0 \in T_n$  and  $T_n$  is bounded above,  $T_n$  has a greatest element by [Theorem 1.3.3](#), which implies that  $(a_n)_{n=1}^{\infty}$  is well-defined. Let  $P(n)$  be the statement that

$$P(n) : 0 \leq t_n < b \text{ and } a_n \text{ is well-defined.}$$

- **Base case:**  $P(1)$  is true since  $0 \leq t_1 = by < b$ .
- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ . Then we have

$$\begin{aligned} t_{n+1} &= b^{n+1}y - \sum_{k=1}^n a_k b^{n+1-k} \\ &= b \left( b^n y - \sum_{k=1}^n a_k b^{n-k} \right) \\ &= b \left( b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} - a_n \right) \\ &= b(t_n - a_n). \end{aligned}$$

Since  $a_n = \max T_n$  and  $t_n < b$ , we have

$$a_n \leq t_n < a_n + 1 \implies 0 \leq t_n - a_n < 1 \implies 0 \leq t_{n+1} = b(t_n - a_n) < b.$$

Thus  $P(n+1)$  is true whenever  $P(n)$  is true.

By the *principle of mathematical induction*,  $P(n)$  is true for all  $n \in \mathbb{N}$ . By the definition of  $a_n$ , we have

$$\begin{aligned} a_n &\leq b^n y - \sum_{k=1}^{n-1} a_k b^{n-k} < a_n + 1 \\ \implies 0 &\leq b^n y - \sum_{k=1}^n a_k b^{n-k} < 1 \\ \implies 0 &\leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \end{aligned}$$

for each  $n \in \mathbb{N}$ . Therefore, the *existence* of the *greedy base  $b$  expansion* of  $y$  is guaranteed.

Next, assume that there exist two sequences  $(a_n)_{n=1}^\infty$  and  $(c_n)_{n=1}^\infty$  such that for each  $n \in \mathbb{N}$ , we have

$$\begin{aligned} \bullet \quad &0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \\ \bullet \quad &0 \leq y - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n} \end{aligned}$$

where  $a_k \in S_b$  and  $c_k \in S_b$  for all  $k = 1, 2, \dots, n$ . We will show that for each  $n \in \mathbb{N}$ , we have  $a_n = c_n$ . Let  $Q(n)$  be the statement that

$$Q(n) : a_k = c_k \text{ for all } k = 1, 2, \dots, n.$$

• **Base case:** We have

$$\begin{aligned} - \quad &0 \leq y - \frac{a_1}{b} < \frac{1}{b} \implies \frac{a_1}{b} \leq y < \frac{a_1 + 1}{b}. \\ - \quad &0 \leq y - \frac{c_1}{b} < \frac{1}{b} \implies \frac{c_1}{b} \leq y < \frac{c_1 + 1}{b}. \end{aligned}$$

By combining the two inequalities, we have

$$\begin{aligned} - \quad &\frac{a_1}{b} \leq y < \frac{c_1 + 1}{b} \implies a_1 < c_1 + 1 \implies a_1 \leq c_1. \\ - \quad &\frac{c_1}{b} \leq y < \frac{a_1 + 1}{b} \implies c_1 < a_1 + 1 \implies c_1 \leq a_1. \end{aligned}$$

Therefore, we have  $a_1 = c_1$ , which shows that  $Q(1)$  is true.

• **Inductive step:** Assume that  $Q(n)$  is true for some  $n \in \mathbb{N}$ . We have

$$- \quad 0 \leq y - \sum_{k=1}^{n+1} \frac{a_k}{b^k} < \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}}.$$

$$- 0 \leq y - \sum_{k=1}^{n+1} \frac{c_k}{b^k} < \frac{1}{b^{n+1}} \implies \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}}.$$

By combining the two inequalities, we have

$$\begin{aligned} - \sum_{k=1}^{n+1} \frac{a_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{c_k}{b^k} + \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{a_k - c_k}{b^k} < \frac{1}{b^{n+1}}. \\ - \sum_{k=1}^{n+1} \frac{c_k}{b^k} \leq y < \sum_{k=1}^{n+1} \frac{a_k}{b^k} + \frac{1}{b^{n+1}} &\implies \sum_{k=1}^{n+1} \frac{c_k - a_k}{b^k} < \frac{1}{b^{n+1}}. \end{aligned}$$

By the inductive hypothesis, we have  $a_k = c_k$  for all  $k = 1, 2, \dots, n$ . Therefore, we have

$$\begin{aligned} - \frac{a_{n+1} - c_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} &\implies a_{n+1} - c_{n+1} < 1 \implies a_{n+1} \leq c_{n+1}. \\ - \frac{c_{n+1} - a_{n+1}}{b^{n+1}} < \frac{1}{b^{n+1}} &\implies c_{n+1} - a_{n+1} < 1 \implies c_{n+1} \leq a_{n+1}. \end{aligned}$$

Therefore, we have  $a_{n+1} = c_{n+1}$ , which shows that  $Q(n+1)$  is true whenever  $Q(n)$  is true.

By the *principle of mathematical induction*,  $Q(n)$  is true for all  $n \in \mathbb{N}$ . Therefore, the *uniqueness* of the *greedy base  $b$  expansion* of  $y$  is guaranteed.  $\square$

**Theorem 1.5.9.** Let  $b \in \mathbb{N}$  be given with  $b \geq 2$ . Let  $(a_n)_{n=1}^\infty$  be the greedy base  $b$  expansion of  $a \in [0, 1)$ , and let  $(c_n)_{n=1}^\infty$  be the greedy base  $b$  expansion of  $c \in [0, 1)$ . If  $a_n = c_n$  for all  $n \in \mathbb{N}$ , then we have  $a = c$ .

*Proof.* Assume that  $a \neq c$ . Without loss of generality, assume that  $a < c$ . Then, by [Corollary 1.5.2](#), there exists some  $n \in \mathbb{N}$  such that

$$\frac{1}{b^n} < c - a.$$

By the definition of the *greedy base  $b$  expansion*, we have

$$\begin{aligned} \bullet \quad 0 \leq a - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} &\implies \sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n}. \\ \bullet \quad 0 \leq c - \sum_{k=1}^n \frac{c_k}{b^k} < \frac{1}{b^n} &\implies c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}. \end{aligned}$$

By the assumption that  $a_n = c_n$  for all  $n \in \mathbb{N}$ , we have

$$\sum_{k=1}^n \frac{a_k}{b^k} = \sum_{k=1}^n \frac{c_k}{b^k}.$$

Therefore, we have a contradiction:

$$\sum_{k=1}^n \frac{a_k}{b^k} + \frac{1}{b^n} \leq a + \frac{1}{b^n} < c < \sum_{k=1}^n \frac{c_k}{b^k} + \frac{1}{b^n}.$$

Therefore, we have  $a = c$ , which completes the proof.  $\square$

**Theorem 1.5.10.** *Let  $b \in \mathbb{N}$  be given with  $b \geq 2$ , and let  $(a_n)_{n=1}^\infty$  be given with  $a_k \in S_b$  for all  $k \in \mathbb{N}$ . If there exists no  $y \in [0, 1)$  such that*

$$(a_n)_{n=1}^\infty \text{ is the greedy base } b \text{ expansion of } y,$$

*then there exists some  $N \in \mathbb{N}$  such that*

$$a_n = b - 1 \quad \text{for all } n \geq N.$$

*Proof.* We will prove the contrapositive of the statement: For all  $N \in \mathbb{N}$ , if there exists some  $n \in \mathbb{N}$  such that

$$n \geq N \text{ and } a_n \neq b - 1,$$

then there exists  $y \in [0, 1)$  such that  $(a_n)_{n=1}^\infty$  is the greedy base  $b$  expansion of  $y$ .

Assume that the condition of the above claim holds. Let  $x_n := \sum_{k=1}^n \frac{a_k}{b^k}$  for each  $n \in \mathbb{N}$ . Since  $a_k \in S_b$  for all  $k \in \mathbb{N}$ , we have

$$0 \leq x_n \leq \sum_{k=1}^n \frac{b-1}{b^k} = 1 - \frac{1}{b^n} < 1 \quad \text{for all } n \in \mathbb{N}.$$

Then, we have that  $X := \{x_n \mid n \in \mathbb{N}\}$  is bounded above. By the *least upper bound property*, there exists  $y = \sup X \geq 0$ .

First, we will show that  $y < 1$ . By the assumption, there exists  $m \in \mathbb{N}$  such that  $a_m \neq b - 1$ . Then, for all  $n > m$ , we have

$$\begin{aligned} x_n &= x_m + \sum_{k=m+1}^n \frac{a_k}{b^k} \leq x_m + \sum_{k=m+1}^n \frac{b-1}{b^k} \\ \implies x_n &\leq x_m + \frac{1}{b^m} - \frac{1}{b^n} < x_m + \frac{1}{b^m}. \end{aligned}$$

Since  $a_m \leq b - 2$ , we have

$$\begin{aligned} x_m + \frac{1}{b^m} &= \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{a_m + 1}{b^m} \\ &\leq \sum_{k=1}^{m-1} \frac{a_k}{b^k} + \frac{b-1}{b^m} \\ &\leq \sum_{k=1}^m \frac{b-1}{b^k} = 1 - \frac{1}{b^m}. \end{aligned}$$

We have  $x_n \leq x_m$  for all  $n \leq m$ , and  $x_n < x_m + \frac{1}{b^m} \leq 1 - \frac{1}{b^m}$  for all  $n > m$ .

Thus  $1 - \frac{1}{b^m}$  is an upper bound of  $X$ . Since  $y = \sup X$ , we have

$$y \leq 1 - \frac{1}{b^m} < 1 \implies y \in [0, 1).$$

Next, we will show that  $(a_n)_{n=1}^\infty$  is the *greedy base  $b$  expansion* of  $y$ . Let  $n \in \mathbb{N}$  be fixed. By the assumption, there exists  $m \in \mathbb{N}$  such that  $m \geq n$  and  $a_m \leq b - 2$ .

$$\begin{aligned} x_m &= x_n + \sum_{k=n+1}^m \frac{a_k}{b^k} < x_n + \sum_{k=n+1}^m \frac{b-1}{b^k} \\ \implies x_m &< x_n + \frac{1}{b^n} - \frac{1}{b^m} \implies x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n}. \end{aligned}$$

Since  $a_m \leq b - 2$ ,  $x_m + \frac{1}{b^m}$  is an upper bound of  $X$  by the same argument as above. Since  $y = \sup X$ , we have

$$y \leq x_m + \frac{1}{b^m} < x_n + \frac{1}{b^n} \implies 0 \leq y - x_n < \frac{1}{b^n}.$$

Therefore,  $(a_n)_{n=1}^\infty$  is the *greedy base  $b$  expansion* of  $y$ . □

**Theorem 1.5.11.** *Let  $b \in \mathbb{N}$  be given with  $b \geq 2$  and let  $(a_n)_{n=1}^\infty$  be given with  $a_k \in S_b$  for all  $k \in \mathbb{N}$ . If there exists some  $N \in \mathbb{N}$  such that*

$$a_n = b - 1 \quad \text{for all } n \geq N,$$

*then there exists no  $y \in [0, 1)$  such that  $(a_n)_{n=1}^\infty$  is the greedy base  $b$  expansion of  $y$ . This theorem is the converse of [Theorem 1.5.10](#).*

*Proof.* Assume that there exists some  $y \in [0, 1)$  such that  $(a_n)_{n=1}^\infty$  is the *greedy base  $b$  expansion* of  $y$ . Then we have

$$0 \leq y - \sum_{k=1}^n \frac{a_k}{b^k} < \frac{1}{b^n} \quad \text{for all } n \in \mathbb{N}.$$

Let  $N \in \mathbb{N}$  be such that  $a_n = b - 1$  for all  $n \geq N$ . Let  $x_n := \sum_{k=1}^n \frac{a_k}{b^k}$  for each  $n \in \mathbb{N}$ . Then, for all  $n \geq N$ , we have

$$\begin{aligned} x_n &= x_{N-1} + \sum_{k=N}^n \frac{b-1}{b^k} = x_{N-1} + \frac{1}{b^{N-1}} - \frac{1}{b^n} \\ \implies y - x_n &= y - x_{N-1} - \frac{1}{b^{N-1}} + \frac{1}{b^n} \geq 0 \\ \implies \frac{1}{b^n} &\geq x_{N-1} + \frac{1}{b^{N-1}} - y. \end{aligned}$$

Since  $0 \leq y - x_{N-1} < \frac{1}{b^{N-1}}$ , we have  $x_{N-1} + \frac{1}{b^{N-1}} - y > 0$ . By [Corollary 1.5.2](#), there exists some  $n \in \mathbb{N}$  with  $n \geq N$  such that

$$\frac{1}{b^n} < x_{N-1} + \frac{1}{b^{N-1}} - y,$$

which is a contradiction. Therefore, there exists no  $y \in [0, 1)$  such that  $(a_n)_{n=1}^\infty$  is the *greedy base  $b$  expansion* of  $y$ .  $\square$

## 1.6 The Extended Real Number System

The Extended real number system  $\overline{\mathbb{R}}$  is defined as

$$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, +\infty\}.$$

We define the following operations on  $\overline{\mathbb{R}}$ :

- $x + (+\infty) := +\infty$  for all  $x \in \mathbb{R} \cup \{+\infty\}$ .
- $x + (-\infty) := -\infty$  for all  $x \in \mathbb{R} \cup \{-\infty\}$ .
- $x \cdot (\pm\infty) := \pm\infty$  for all  $x \in \mathbb{R}^+ \cup \{+\infty\}$ .
- $x \cdot (\pm\infty) := \mp\infty$  for all  $x \in \mathbb{R}^- \cup \{-\infty\}$ .
- $\frac{x}{\pm\infty} := 0$  for all  $x \in \mathbb{R}$ .

The following operations are undefined in  $\overline{\mathbb{R}}$ :

- $(\pm\infty) + (\mp\infty)$
- $0 \cdot (\pm\infty)$
- $\frac{(\pm\infty)}{(\pm\infty)}$  and  $\frac{(\pm\infty)}{(\mp\infty)}$
- $\frac{x}{0}$  for all  $x \in \overline{\mathbb{R}}$

$\overline{\mathbb{R}}$  is not a field, but it is an ordered set with the following order relation:

$$-\infty < x < +\infty \quad \text{for all } x \in \mathbb{R}.$$

*Remark.* We can identify the following intervals in  $\overline{\mathbb{R}}$ :

- $(-\infty, a) = \{x \in \mathbb{R} \mid x < a\}$  for all  $a \in \mathbb{R}$ .
- $(a, +\infty) = \{x \in \mathbb{R} \mid x > a\}$  for all  $a \in \mathbb{R}$ .
- $(-\infty, +\infty) = \mathbb{R}$ .

**Theorem 1.6.1.** Any subset of  $\overline{\mathbb{R}}$  has a least upper bound and a greatest lower bound in  $\overline{\mathbb{R}}$ .

*Proof.* Let  $E \subseteq \overline{\mathbb{R}}$ . If  $E = \emptyset$ , then we have

- $\sup E = -\infty$
- $\inf E = +\infty$ .

Otherwise, we have the following cases:

- If  $E$  is bounded above in  $\mathbb{R}$ , then  $\sup E$  exists in  $\mathbb{R}$  by the *least upper bound property* of  $\mathbb{R}$ .
- If  $E$  is not bounded above in  $\mathbb{R}$ , then  $\sup E = +\infty$ .
- If  $E$  is bounded below in  $\mathbb{R}$ , then  $\inf E$  exists in  $\mathbb{R}$  by [Theorem 1.3.7](#).
- If  $E$  is not bounded below in  $\mathbb{R}$ , then  $\inf E = -\infty$ .

□

## 1.7 The Complex Field

The complex field  $\mathbb{C}$  is defined as

$$\mathbb{C} := \{(x, y) \mid x, y \in \mathbb{R}\}.$$

We define the following operations on  $\mathbb{C}$ :

- $(x, y) + (u, v) := (x + u, y + v)$  for all  $(x, y), (u, v) \in \mathbb{C}$ .
- $(x, y) \cdot (u, v) := (xu - yv, xv + yu)$  for all  $(x, y), (u, v) \in \mathbb{C}$ .

$\mathbb{C}$  is a field with the following properties:

- The additive identity is  $(0, 0)$  and the multiplicative identity is  $(1, 0)$ .
- The additive inverse of  $(x, y)$  is  $(-x, -y)$  and the multiplicative inverse of  $(x, y)$  is

$$\left( \frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right)$$

for all  $(x, y) \in \mathbb{C}$  with  $(x, y) \neq (0, 0)$ .

*Remark.* The function

$$f : \{(x, 0) \mid x \in \mathbb{R}\} \subseteq \mathbb{C} \rightarrow \mathbb{R}$$

defined by  $f((x, 0)) = x$  is an *isomorphism*. We can identify  $\mathbb{R}$  with the subfield  $\{(x, 0) \mid x \in \mathbb{R}\}$  of  $\mathbb{C}$ . We denote  $(x, 0) \in \mathbb{C}$  by  $x \in \mathbb{R}$ .

*Remark.* We define  $i := (0, 1) \in \mathbb{C}$ . Then we have  $i^2 = -1$ , and any complex number  $(x, y) \in \mathbb{C}$  can be written as

$$(x, y) = x + yi.$$

**Definition 1.7.1.** Let  $z = x + yi \in \mathbb{C}$  for some  $x, y \in \mathbb{R}$ . The **real part** of  $z$  is defined as

$$\Re(z) := x \text{ or } \operatorname{Re}(z) := x,$$

the **imaginary part** of  $z$  is defined as

$$\Im(z) := y \text{ or } \operatorname{Im}(z) := y,$$

and the **complex conjugate** of  $z$  is defined as

$$\bar{z} := x - yi.$$

**Theorem 1.7.2.** Let  $z = x + yi \in \mathbb{C}$  and let  $w = u + vi \in \mathbb{C}$  for some  $x, y, u, v \in \mathbb{R}$ . Then we have the following properties:

- $\overline{(\overline{z})} = z$ .
- $z = \overline{z} \iff z \in \mathbb{R}$ .
- $\overline{z + w} = \overline{z} + \overline{w}$ .
- $\overline{zw} = \overline{z} \cdot \overline{w}$ .
- $z + \overline{z} = 2\operatorname{Re}(z)$  and  $z - \overline{z} = 2i\operatorname{Im}(z)$ .
- $z\overline{z} = x^2 + y^2 \in \mathbb{R}^+ \cup \{0\}$ .

*Proof.* The proof follows directly from the definitions and properties of the operations on  $\mathbb{C}$ .  $\square$

**Definition 1.7.3.** Let  $z = x + yi \in \mathbb{C}$  for some  $x, y \in \mathbb{R}$ . the **modulus** of  $z$  is defined as

$$|z| := \sqrt{z\overline{z}} = \sqrt{x^2 + y^2} \in \mathbb{R}^+ \cup \{0\}.$$

The *existence* and *uniqueness* of  $|z|$  follows from [Theorems 1.5.6](#) and [1.7.2](#).

**Theorem 1.7.4.** Let  $a_1, a_2, \dots, a_n \in \mathbb{C}$  and  $b_1, b_2, \dots, b_n \in \mathbb{C}$ . Then we have

$$\left| \sum_{k=1}^n a_k \overline{b_k} \right|^2 \leq \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2.$$

This theorem is called the **Cauchy-Schwarz inequality**.

*Proof.* Let  $i, j \in \mathbb{N}$  with  $1 \leq i < j \leq n$ . Then we have

$$\begin{aligned} |a_i b_j - a_j b_i|^2 &= (a_i b_j - a_j b_i) \overline{(a_i b_j - a_j b_i)} \\ &= |a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2 - (a_i \overline{b_i} \cdot \overline{a_j} b_j + \overline{a_i} b_i \cdot a_j \overline{b_j}). \end{aligned}$$

Consider the following sums:

$$\begin{aligned} \sum_{i < j} (|a_i|^2 |b_j|^2 + |a_j|^2 |b_i|^2) &= \sum_{i < j} (|a_i|^2 |b_j|^2) + \sum_{j < i} (|a_i|^2 |b_j|^2) \\ &= \sum_{i \neq j} (|a_i|^2 |b_j|^2) \\ &= \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2. \end{aligned}$$

$$\begin{aligned}
\sum_{i < j} (a_i \bar{b}_i \cdot \bar{a}_j b_j + \bar{a}_i b_i \cdot a_j \bar{b}_j) &= \sum_{i \neq j} (a_i \bar{b}_i \cdot \bar{a}_j b_j) \\
&= \sum_{i=1}^n a_i \bar{b}_i \cdot \sum_{j=1}^n \bar{a}_j b_j - \sum_{k=1}^n a_k \bar{b}_k \cdot \bar{a}_k b_k \\
&= \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 - \sum_{k=1}^n |a_k|^2 |b_k|^2.
\end{aligned}$$

Therefore, we have

$$\sum_{i < j} |a_i b_j - a_j b_i|^2 = \sum_{k=1}^n |a_k|^2 \cdot \sum_{k=1}^n |b_k|^2 - \left| \sum_{k=1}^n a_k \bar{b}_k \right|^2 \geq 0.$$

□

**Corollary 1.7.5.** Let  $a_1, a_2, \dots, a_n \in \mathbb{R}$  and  $b_1, b_2, \dots, b_n \in \mathbb{R}$ . Then we have

$$\left( \sum_{k=1}^n a_k b_k \right)^2 \leq \sum_{k=1}^n a_k^2 \cdot \sum_{k=1}^n b_k^2.$$

**Theorem 1.7.6.** Let  $z, w \in \mathbb{C}$ . Then we have the following properties:

- (A)  $|zw| = |z| \cdot |w|$ .
- (B)  $|z + w| \leq |z| + |w|$ .
- (C)  $||z| - |w|| \leq |z - w|$ .

*Proof.* Let  $z = x + yi$  and  $w = u + vi$  for some  $x, y, u, v \in \mathbb{R}$ .

- (A)  $|zw|^2 = (xu - yv)^2 + (xv + yu)^2 = (x^2 + y^2)(u^2 + v^2) = |z|^2 \cdot |w|^2$ .
- (B) By [Corollary 1.7.5](#), we have  $(xu + yv)^2 \leq (x^2 + y^2)(u^2 + v^2)$ .

$$\begin{aligned}
|z + w|^2 &= (x + u)^2 + (y + v)^2 \\
&= x^2 + y^2 + u^2 + v^2 + 2(xu + yv) \\
&\leq x^2 + y^2 + u^2 + v^2 + 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\
&= (|z| + |w|)^2.
\end{aligned}$$

- (C) Again, by [Corollary 1.7.5](#), we have

$$\begin{aligned}
|z - w|^2 &= (x - u)^2 + (y - v)^2 \\
&= x^2 + y^2 + u^2 + v^2 - 2(xu + yv) \\
&\geq x^2 + y^2 + u^2 + v^2 - 2\sqrt{x^2 + y^2} \cdot \sqrt{u^2 + v^2} \\
&= (|z| - |w|)^2.
\end{aligned}$$

□

## 1.8 Euclidean Spaces

For each  $n \in \mathbb{N}$ , we call the set of all ordered  $n$ -tuples of real numbers as the **Euclidean  $n$ -space**, and denote it as  $\mathbb{R}^n$ . We can identify  $\mathbb{R}^n$  as

$$\mathbb{R}^n = \{\mathbf{x} = (x_1, x_2, \dots, x_n) \mid x_k \in \mathbb{R} \text{ for all } k = 1, 2, \dots, n\},$$

where  $x_1, x_2, \dots, x_n$  are called the **coordinates** of  $\mathbf{x}$ . The elements of  $\mathbb{R}^n$  with  $n > 1$  are called **points** (or *vectors*), which are denoted by boldface letters.

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  for some  $n \in \mathbb{N}$ , and let  $\alpha \in \mathbb{R}$ . We define the following operations on  $\mathbb{R}^n$ :

- $\mathbf{x} + \mathbf{y} := (x_1 + y_1, x_2 + y_2, \dots, x_n + y_n)$ .
- $\alpha \mathbf{x} := (\alpha x_1, \alpha x_2, \dots, \alpha x_n)$ .

Then,  $\mathbb{R}^n$  is a *vector space* over  $\mathbb{R}$  with the zero vector  $\mathbf{0} = (0, 0, \dots, 0)$ .

**Definition 1.8.1.** Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  for some  $n \in \mathbb{N}$ . The **inner product** of  $\mathbf{x}$  and  $\mathbf{y}$  is defined as

$$\mathbf{x} \cdot \mathbf{y} := \sum_{k=1}^n x_k y_k.$$

The **norm** of  $\mathbf{x}$  is defined as

$$\|\mathbf{x}\| := \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{k=1}^n x_k^2}.$$

The **distance** between  $\mathbf{x}$  and  $\mathbf{y}$  is defined as

$$d(\mathbf{x}, \mathbf{y}) := \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}.$$

We can confirm that  $d(x, y) = |x - y|$  in the case of  $n = 1$ .

**Theorem 1.8.2.** Let  $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$  for some  $n \in \mathbb{N}$ . Then we have the following properties:

- (A)  $|\mathbf{x} \cdot \mathbf{y}| \leq \|\mathbf{x}\| \cdot \|\mathbf{y}\|$ .
- (B)  $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ .
- (C)  $\|\mathbf{x} - \mathbf{y}\| \geq \|\mathbf{x}\| - \|\mathbf{y}\|$ .

*Proof.* (A) By [Corollary 1.7.5](#), we have

$$|\mathbf{x} \cdot \mathbf{y}|^2 = \left| \sum_{k=1}^n x_k y_k \right|^2 \leq \sum_{k=1}^n x_k^2 \cdot \sum_{k=1}^n y_k^2 = \|\mathbf{x}\|^2 \cdot \|\mathbf{y}\|^2.$$

(B) By part (A), we have

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\|^2 &= (\mathbf{x} + \mathbf{y}) \cdot (\mathbf{x} + \mathbf{y}) = \|\mathbf{x}\|^2 + 2\mathbf{x} \cdot \mathbf{y} + \|\mathbf{y}\|^2 \\ &\leq \|\mathbf{x}\|^2 + 2|\mathbf{x} \cdot \mathbf{y}| + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\| \cdot \|\mathbf{y}\| + \|\mathbf{y}\|^2 \\ &= (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \end{aligned}$$

(C) By part (B), we have

$$\begin{aligned} \|\mathbf{x}\| &= \|(\mathbf{x} - \mathbf{y}) + \mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y}\| \implies \|\mathbf{x}\| - \|\mathbf{y}\| \leq \|\mathbf{x} - \mathbf{y}\|. \\ \|\mathbf{y}\| &= \|(\mathbf{y} - \mathbf{x}) + \mathbf{x}\| \leq \|\mathbf{y} - \mathbf{x}\| + \|\mathbf{x}\| \implies \|\mathbf{y}\| - \|\mathbf{x}\| \leq \|\mathbf{x} - \mathbf{y}\|. \end{aligned}$$

□

## Chapter 2

# Basic Topology

### 2.1 Functions

A *function* (or a *mapping*) is defined with 3 components.

- A **domain**  $X$ .
- A **codomain**  $Y$ .
- A **rule**  $f$  which associates each element  $x \in X$  to a *unique* element  $y \in Y$ .

$f$  is said to be a *function* of  $X$  into  $Y$ , which is denoted by  $f : X \rightarrow Y$ . For each  $x \in X$ , the *unique* element  $y \in Y$  associated with  $x$  is denoted by  $f(x)$ .

*Remark.* Let  $f : X \rightarrow Y$  be a function. For any subset  $E \subseteq X$ , we define the **image** of  $E$  under  $f$  as

$$f(E) := \{f(x) \mid x \in E\} \subseteq Y.$$

Especially, if  $E = X$ , then  $f(X) \subseteq Y$  is called the **range** of  $f$ .

*Remark.* Let  $f : X \rightarrow Y$  be a function. For any subset  $E \subseteq Y$ , we define the **inverse image** of  $E$  under  $f$  as

$$f^{-1}(E) := \{x \in X \mid f(x) \in E\} \subseteq X.$$

**Definition 2.1.1.** Let  $f : X \rightarrow Y$  be a function. We define the properties of  $f$  as follows:

- $f$  is **injective** (or *one-to-one*) if for any  $x_1, x_2 \in X$ , we have

$$f(x_1) = f(x_2) \implies x_1 = x_2.$$

- $f$  is **surjective** (or *onto*) if  $f(X) = Y$ .
- $f$  is **bijective** (or a *one-to-one correspondence*) if it is both injective and surjective.

**Definition 2.1.2.** Let  $f : X \rightarrow Y$  be a bijective function. Then, for any  $y \in Y$ , there exists a *unique*  $x \in X$  such that  $f(x) = y$ . Therefore, we can define a function

$$f^{-1} : Y \rightarrow X \quad \text{such that} \quad f^{-1}(y) = x.$$

Such a function  $f^{-1}$  is called the **inverse function** of  $f$ .

**Theorem 2.1.3.** Let  $f : X \rightarrow Y$  be a bijective function. Then the inverse function  $f^{-1} : Y \rightarrow X$  is bijective.

*Proof.* Let  $y_1, y_2 \in Y$  be given. Assume that  $f^{-1}(y_1) = f^{-1}(y_2) = x$ . Then, by [Definition 2.1.2](#), we have

$$f(x) = y_1 \text{ and } f(x) = y_2 \implies y_1 = y_2$$

since  $f$  is a function. Therefore, we have that  $f^{-1}$  is injective. By the definition of image, we have that  $f^{-1}(Y) = X$ , which implies that  $f^{-1}$  is surjective.  $\square$

**Definition 2.1.4.** Let  $f : X \rightarrow Y$  be a function. Let  $E \subseteq X$  and  $F \subseteq Y$  be given with  $f(E) \subseteq F$ . We define the restriction functions as follows:

- The **restriction** of  $f$  to  $E$  is defined by

$$f|_E : E \rightarrow Y \quad \text{such that} \quad f|_E(x) = f(x) \quad \text{for all } x \in E.$$

- The **Corestriction** of  $f$  to  $F$  is defined by

$$f|_F : X \rightarrow F \quad \text{such that} \quad f|_F(x) = f(x) \quad \text{for all } x \in X.$$

- The **Bi-restriction** of  $f$  to  $E$  and  $F$  is defined by

$$f|_E^F : E \rightarrow F \quad \text{such that} \quad f|_E^F(x) = f(x) \quad \text{for all } x \in E.$$

We will simply call all of them as just *restriction*, if there is no ambiguity.

**Theorem 2.1.5.** Let  $f : X \rightarrow Y$  be a function. Let  $E \subseteq X$  and  $F \subseteq Y$  be given with  $f(E) \subseteq F$ . Then we have the following properties:

(A) If  $f$  is injective, then  $f|_E^F$  is also injective.

(B) If  $f(E) = F$ , then  $f|_E^F$  is surjective.

(C) If  $f$  is injective and  $f(E) = F$ , then  $f|_E^F$  is bijective.

*Proof.* (C) follows from (A) and (B). Let  $x_1, x_2 \in E$ .

(A) Assume that  $f$  is injective. Then we have

$$f|_E^F(x_1) = f|_E^F(x_2) \implies f(x_1) = f(x_2) \implies x_1 = x_2.$$

(B) Assume that  $f(E) = F$ . Then, for any  $y \in F$ , there exists  $x \in E$  such that  $f(x) = y$ . Therefore, we have  $f|_E^F(E) = F$ .

□

**Definition 2.1.6.** Let  $X, Y, Z$  be sets and  $f : X \rightarrow Y, g : Y \rightarrow Z$  be functions. We define the **composition** of  $f$  and  $g$  as the function  $g \circ f : X \rightarrow Z$  defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in X.$$

**Theorem 2.1.7.** Let  $X, Y, Z$  be sets and  $f : X \rightarrow Y, g : Y \rightarrow Z$  be functions. Then we have the following properties:

(A) If  $f$  and  $g$  are injective, then  $g \circ f$  is also injective.

(B) If  $f$  and  $g$  are surjective, then  $g \circ f$  is also surjective.

(C) If  $f$  and  $g$  are bijective, then  $g \circ f$  is also bijective.

*Proof.* (C) follows from (A) and (B). Let  $x_1, x_2 \in X$  be given.

(A) Assume that  $f$  and  $g$  are injective. Then we have

$$\begin{aligned} (g \circ f)(x_1) = (g \circ f)(x_2) &\implies g(f(x_1)) = g(f(x_2)) \\ &\implies f(x_1) = f(x_2) \implies x_1 = x_2. \end{aligned}$$

(B) Assume that  $f$  and  $g$  are surjective. Then, for any  $z \in Z$ , there exists  $y \in Y$  such that  $g(y) = z$ . Since  $f$  is surjective, there exists  $x \in X$  such that  $f(x) = y$ . Therefore, we have  $(g \circ f)(x) = g(f(x)) = g(y) = z$ .

□

**Theorem 2.1.8.** *Let  $S$  be a set of sets. For [Sections 2.2](#) and [2.3](#), we consider the relation  $\sim$  defined on  $S$  by*

$$X \sim Y \iff \text{There exists a bijection } f : X \rightarrow Y.$$

*The relation  $\sim$  is an equivalence relation on  $S$ .*

*Proof.* Let  $X, Y, Z \in S$  be given.

- **Reflexivity:** The identity function  $id_X : X \rightarrow X$ , defined by  $id_X(x) = x$  for all  $x \in X$ , is bijective. Therefore, we have  $X \sim X$ .
- **Symmetry:** Assume that  $X \sim Y$ . Then there exists a bijection  $f : X \rightarrow Y$ . By [Theorem 2.1.3](#), the inverse function  $f^{-1} : Y \rightarrow X$  is also bijective. Therefore, we have  $Y \sim X$ .
- **Transitivity:** Assume that  $X \sim Y$  and  $Y \sim Z$ . Then there exist bijections  $f : X \rightarrow Y$  and  $g : Y \rightarrow Z$ . By [Theorem 2.1.7](#), the composition  $g \circ f : X \rightarrow Z$  is also bijective. Therefore, we have  $X \sim Z$ .

□

## 2.2 Finite Sets

Let  $J_n$  be a set defined by

$$J_n := \{1, 2, \dots, n\} \quad \text{for each } n \in \mathbb{N}.$$

A set  $X$  is said to be **finite** if  $J_n \sim X$  for some  $n \in \mathbb{N}$ . In this case,  $n$  is called the **cardinality** of  $X$ , denoted by  $|X|$ . Especially, if  $X = \emptyset$ , we define  $|X| = 0$ .  $X$  is said to be **infinite** if  $X$  is not finite.

**Lemma 2.2.1.** *Any subset of  $J_n$  is finite for each  $n \in \mathbb{N}$ .*

*Proof.* Let  $P(n)$  be the statement that any subset of  $J_n$  is finite.

- **Base case:**  $J_1 = \{1\}$  has  $\emptyset$  and  $J_1$  as its only subsets. We have  $J_1 \sim J_1$  by  $f \equiv id_{J_1}$ , and  $\emptyset$  is finite by definition. Thus  $P(1)$  is true.
- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ . Let  $A \subseteq J_{n+1}$  be given. We have two cases:
  - If  $n+1 \notin A$ , then  $A \subseteq J_n$ , which is finite by the inductive hypothesis.
  - If  $n+1 \in A$ , then  $A = B \cup \{n+1\}$  for some  $B \subseteq J_n$ . By the inductive hypothesis,  $B$  is finite, so there exists a bijection  $f : J_m \rightarrow B$  for some  $m \in \mathbb{N}$ . Let  $g : J_{m+1} \rightarrow A$  be defined by

$$g(k) = \begin{cases} f(k) & \text{if } k \leq m \\ n+1 & \text{if } k = m+1 \end{cases}$$

Then  $g$  is bijective, which implies that  $A$  is finite.

Thus  $P(n + 1)$  is true whenever  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ .  $\square$

**Theorem 2.2.2.** *Any subset of a finite set is finite.*

*Proof.* Let  $X$  be a finite set and  $Y \subseteq X$ . Then there exists a bijection  $f : J_n \rightarrow X$  for some  $n \in \mathbb{N}$ . Let  $Z$  be defined by

$$Z := f^{-1}(Y) \subseteq J_n.$$

Since  $Z \subseteq J_n$ ,  $Z$  is finite by Lemma 2.2.1, so there exists a bijection  $g : J_m \rightarrow Z$  for some  $m \in \mathbb{N}$ .

By Theorem 2.1.5,  $f|_Z^Y$  is bijective. Then, by Theorem 2.1.7, the composition  $f|_Z^Y \circ g : J_m \rightarrow Y$  is also bijective. Therefore,  $Y$  is finite.  $\square$

**Lemma 2.2.3.** *Let  $J_n$  be given for some  $n \in \mathbb{N}$ . If there exists an surjection  $f : J_n \rightarrow Y$ , then  $Y$  is finite.*

*Proof.* Let  $f : J_n \rightarrow Y$  be a surjective function. For any  $y \in Y$ , let  $N_y$  be a set defined by

$$N_y := f^{-1}(\{y\}) \subseteq J_n.$$

Since  $f$  is surjective,  $N_y$  is nonempty. By Theorem 1.1.3,  $N_y$  has a least element, say  $n_y$ . Let  $M_Y$  be a set defined by

$$M_Y := \{n_y \mid y \in Y\} \subseteq J_n.$$

Let  $g : M_Y \rightarrow Y$  be a function defined by

$$g(n_y) = y \quad \text{for each } n_y \in M_Y.$$

Then  $g$  is bijective. By Lemma 2.2.1,  $M_Y$  is finite, so there exists a bijection  $h : J_m \rightarrow M_Y$  for some  $m \in \mathbb{N}$ . Then, by Theorem 2.1.7, the composition  $g \circ h : J_m \rightarrow Y$  is bijective. Therefore,  $Y$  is finite.  $\square$

**Theorem 2.2.4.** *Let  $X$  be a finite set. If there exists an surjection  $f : X \rightarrow Y$ , then  $Y$  is finite.*

*Proof.* Let  $f : X \rightarrow Y$  be a surjective function. Since  $X$  is finite, there exists a bijection  $g : J_n \rightarrow X$  for some  $n \in \mathbb{N}$ . Then, by Theorem 2.1.7, the composition  $f \circ g : J_n \rightarrow Y$  is surjective. Therefore, by Lemma 2.2.3,  $Y$  is finite.  $\square$

**Theorem 2.2.5.** *Let  $X$  be a finite nonempty ordered set. Then  $X$  has both a least element and a greatest element.*

*Proof.* Since  $X$  is finite and nonempty, there exists a bijection  $f : J_n \rightarrow X$  for some  $n \in \mathbb{N}$ . The elements of  $X$  can be listed as  $X = \{a_1, a_2, \dots, a_n\}$ , where  $a_k = f(k)$  for each  $k = 1, 2, \dots, n$ .

First, we will show that  $X$  has a least element. Let  $P(n)$  be the statement that any finite ordered set of cardinality  $n$  has a least element.

- **Base case:**  $X = \{a_1\}$  has a least element  $a_1$ . Thus  $P(1)$  is true.
- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ . Let  $X$  be a finite ordered set of cardinality  $n + 1$  such that

$$X = \{a_1, a_2, \dots, a_{n+1}\}.$$

By the inductive hypothesis, a finite ordered subset  $Y \subseteq X$  of cardinality  $n$  such that

$$Y = \{a_1, a_2, \dots, a_n\}$$

has a least element, say  $a_k$  for some  $k = 1, 2, \dots, n$ . We have two cases:

- If  $a_k \leq a_{n+1}$ , then  $a_k$  is the least element of  $X$ .
- If  $a_{n+1} < a_k$ , then  $a_{n+1}$  is the least element of  $X$ .

Thus  $P(n + 1)$  is true whenever  $P(n)$  is true.

By the *principle of mathematical induction*,  $P(n)$  is true for all  $n \in \mathbb{N}$ .

Next, we will show that  $X$  has a greatest element. Let  $Q(n)$  be the statement that any finite ordered set of cardinality  $n$  has a greatest element.

- **Base case:**  $X = \{a_1\}$  has a greatest element  $a_1$ . Thus  $Q(1)$  is true.
- **Inductive step:** Assume that  $Q(n)$  is true for some  $n \in \mathbb{N}$ . Let  $X$  be a finite ordered set of cardinality  $n + 1$  such that

$$X = \{a_1, a_2, \dots, a_{n+1}\}.$$

By the inductive hypothesis, a finite ordered subset  $Y \subseteq X$  of cardinality  $n$  such that

$$Y = \{a_1, a_2, \dots, a_n\}$$

has a greatest element, say  $a_k$  for some  $k = 1, 2, \dots, n$ . We have two cases:

- If  $a_k \geq a_{n+1}$ , then  $a_k$  is the greatest element of  $X$ .
- If  $a_{n+1} > a_k$ , then  $a_{n+1}$  is the greatest element of  $X$ .

Thus  $Q(n+1)$  is true whenever  $Q(n)$  is true.

By the principle of mathematical induction,  $Q(n)$  is true for all  $n \in \mathbb{N}$ .  $\square$

**Lemma 2.2.6.** *Let  $J_m, J_n$  be given for some  $m, n \in \mathbb{N}$ . Then we have the following properties:*

- (A) *If there exists an injection  $f : J_m \rightarrow J_n$ , then we have  $m \leq n$ .*
- (B) *If there exists a surjection  $f : J_m \rightarrow J_n$ , then we have  $m \geq n$ .*
- (C) *If there exists a bijection  $f : J_m \rightarrow J_n$ , then we have  $m = n$ .*

*This lemma guarantees the uniqueness of the cardinality. If  $J_m \sim X$  and  $J_n \sim X$ , then we have  $J_m \sim J_n$ , which implies that  $m = n$  by (C).*

*Proof.* We will prove (A) by mathematical induction on  $m$ , (B) by mathematical induction on  $n$ , and (C) follows from (A) and (B).

- (A) Let  $n \in \mathbb{N}$  be fixed. Let  $P(m)$  be the statement that if there exists an injective function  $f_m : J_m \rightarrow J_n$ , then we have  $m \leq n$ .

- **Base case:** Let  $m = 1$ . Then  $m \leq n$  holds. Thus  $P(1)$  is true.
- **Inductive step:** Assume that  $P(m)$  is true for some  $m \in \mathbb{N}$ . If  $n = 1$ , there does not exist an injective function  $f_{m+1} : J_{m+1} \rightarrow J_n$  since  $f_{m+1}(1) \neq f_{m+1}(m+1)$ , where  $P(m+1)$  is true.

Otherwise, assume that such a function  $f_{m+1}$  exists. We have two cases:

- If  $f_{m+1}(m+1) = n$ , let  $f_m$  be a function defined by

$$f_m \equiv f_{m+1}|_{J_m}^{J_{n-1}} : J_m \rightarrow J_{n-1}.$$

Then  $f_m$  is injective by [Theorem 2.1.5](#). By the inductive hypothesis, we have  $m \leq n-1$ , which implies that  $m+1 \leq n$ .

- If  $f_{m+1}(m+1) = x \neq n$ , let  $g_n : J_n \rightarrow J_n$  be defined by

$$g_n(y) = \begin{cases} x, & \text{if } y = n \\ n, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then  $g_n$  is bijective. Let  $f_m$  be a function defined by

$$f_m \equiv (g_n \circ f_{m+1})|_{J_m}^{J_{n-1}} : J_m \rightarrow J_{n-1}.$$

Then  $f_m$  is injective by [Theorems 2.1.5](#) and [2.1.7](#). By the inductive hypothesis, we have  $m \leq n-1$ , which implies that  $m+1 \leq n$ .

Therefore,  $P(m+1)$  is true whenever  $P(m)$  is true.

By the *principle of mathematical induction*,  $P(m)$  is true for all  $m \in \mathbb{N}$ .

(B) Let  $m \in \mathbb{N}$  be fixed. Let  $Q(n)$  be the statement that if there exists a surjective function  $f_n : J_m \rightarrow J_n$ , then we have  $m \geq n$ .

- **Base case:** Let  $n = 1$ . Then  $m \geq n$  holds. Thus  $Q(1)$  is true.
- **Inductive step:** Assume that  $Q(n)$  is true for some  $n \in \mathbb{N}$ . If  $m = 1$ , there does not exist a surjective function  $f_{n+1} : J_m \rightarrow J_{n+1}$  since  $f_{n+1}(1)$  is unique, where  $Q(n+1)$  is true.

Otherwise, assume that such a function  $f_{n+1}$  exists. We have two cases:

- If  $f_{n+1}(m) = n+1$ , let  $K_{n+1}$  be a set defined by

$$K_{n+1} := (f_{n+1})^{-1}(\{n+1\}) \subseteq J_m.$$

Let  $f_n : J_m \rightarrow J_n$  be a function defined by

$$f_n(y) = \begin{cases} f_{n+1}(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}$$

Then  $f_n$  is surjective. By the inductive hypothesis, we have  $m-1 \geq n$ , which implies that  $m \geq n+1$ .

- If  $f_{n+1}(m) = x \neq n+1$ , let  $g_{n+1} : J_{n+1} \rightarrow J_{n+1}$  be a function defined by

$$g_{n+1}(y) = \begin{cases} x, & \text{if } y = n+1 \\ n+1, & \text{if } y = x \\ y, & \text{otherwise} \end{cases}$$

Then  $g_{n+1}$  is bijective. Let  $K_{n+1}$  be a set defined by

$$K_{n+1} := (g_{n+1} \circ f_{n+1})^{-1}(\{n+1\}) \subseteq J_m.$$

Let  $f_n : J_m \rightarrow J_n$  be a function defined by

$$f_n(y) = \begin{cases} (g_{n+1} \circ f_{n+1})(y), & \text{if } y \in J_m \setminus K_{n+1} \\ n, & \text{if } y \in K_{n+1} \end{cases}$$

Then  $f_n$  is surjective by [Theorem 2.1.7](#). By the inductive hypothesis, we have  $m-1 \geq n$ , which implies that  $m \geq n+1$ .

Thus  $Q(n+1)$  is true whenever  $Q(n)$  is true.

By the *principle of mathematical induction*,  $Q(n)$  is true for all  $n \in \mathbb{N}$ .

□

**Lemma 2.2.7.** *Let  $J_m, J_n$  be given for some  $m, n \in \mathbb{N}$ . Then we have the following properties:*

(A) *If  $m \leq n$ , then there exists an injection  $f : J_m \rightarrow J_n$ .*

(B) *If  $m \geq n$ , then there exists a surjection  $f : J_m \rightarrow J_n$ .*

(C) *If  $m = n$ , then there exists a bijection  $f : J_m \rightarrow J_n$ .*

*This lemma is the converse of [Lemma 2.2.6](#).*

*Proof.* We will prove this lemma by explicitly constructing the functions.

(A) If  $m \leq n$ , let  $f : J_m \rightarrow J_n$  be a function defined by

$$f(i) = i \quad \text{for all } i \in J_m.$$

Then  $f$  is an injection.

(B) If  $m \geq n$ , let  $f : J_m \rightarrow J_n$  be a function defined by

$$f(i) = \begin{cases} i, & \text{if } i \leq n \\ n, & \text{if } i > n \end{cases}$$

Then  $f$  is a surjection.

(C) If  $m = n$ , let  $f : J_m \rightarrow J_n$  be a function defined by

$$f(i) = i \quad \text{for all } i \in J_m.$$

Then  $f$  is a bijection.

□

**Theorem 2.2.8.** *Let  $X, Y$  be finite sets. Then we have the following properties:*

(A) *If there exists an injection  $f : X \rightarrow Y$ , then we have  $|X| \leq |Y|$ .*

(B) *If there exists a surjection  $f : X \rightarrow Y$ , then we have  $|X| \geq |Y|$ .*

(C) *If there exists a bijection  $f : X \rightarrow Y$ , then we have  $|X| = |Y|$ .*

*Moreover, the converses of (A), (B), and (C) are also true.*

*The contraposition of (A) is called the **pigeonhole principle**, which states that if  $|X| > |Y|$ , then there exists no injection  $f : X \rightarrow Y$ .*

*Proof.* Let  $|X| = m$  and  $|Y| = n$  for some  $m, n \in \mathbb{N}$ . Then there exist bijections  $f_X : J_m \rightarrow X$  and  $f_Y : J_n \rightarrow Y$ . By [Theorems 2.1.3](#) and [2.1.7](#), the composition

$$F \equiv (f_Y)^{-1} \circ f \circ f_X : J_m \rightarrow J_n$$

is injective, surjective, or bijective if  $f$  is injective, surjective, or bijective, respectively. Then, by [Lemma 2.2.6](#), we have the desired results:

- (A) If  $f$  is injective, then  $F$  is injective, which implies that  $m \leq n$ .
- (B) If  $f$  is surjective, then  $F$  is surjective, which implies that  $m \geq n$ .
- (C) If  $f$  is bijective, then  $F$  is bijective, which implies that  $m = n$ .

Also, by [Theorems 2.1.3](#) and [2.1.7](#), the composition

$$f \equiv f_Y \circ G \circ (f_X)^{-1} : X \rightarrow Y$$

is injective, surjective, or bijective if  $G : J_m \rightarrow J_n$  is injective, surjective, or bijective, respectively. Then, by [Lemma 2.2.7](#), we have the desired converses:

- (A) If  $m \leq n$ , then there exists an injective function  $G : J_m \rightarrow J_n$ , which implies that there exists an injective function  $f : X \rightarrow Y$ .
- (B) If  $m \geq n$ , then there exists a surjective function  $G : J_m \rightarrow J_n$ , which implies that there exists a surjective function  $f : X \rightarrow Y$ .
- (C) If  $m = n$ , then there exists a bijective function  $G : J_m \rightarrow J_n$ , which implies that there exists a bijective function  $f : X \rightarrow Y$ .

□

**Lemma 2.2.9.** *Let  $X, Y$  be finite sets with  $X \cap Y = \emptyset$ . Then the union  $X \cup Y$  is finite.*

*Proof.* Let  $m = |X|$  and  $n = |Y|$ . There exist bijections  $f : J_m \rightarrow X$  and  $g : J_n \rightarrow Y$ . Let  $h : J_{m+n} \rightarrow X \cup Y$  be a function defined by

$$h(i) = \begin{cases} f(i), & \text{if } i \leq m \\ g(i - m), & \text{if } m < i \leq m + n \end{cases}$$

Then  $h$  is bijective, which implies that  $X \cup Y$  is finite.

□

**Theorem 2.2.10.** *Let  $X, Y$  be finite sets. Then the union  $X \cup Y$  is finite.*

*Proof.* Let  $A, B, C$  be sets defined by

- $A = X \setminus Y \subseteq X$ .
- $B = Y \setminus X \subseteq Y$ .
- $C = X \cap Y \subseteq X$ .

Then  $A, B, C$  are finite by [Theorem 2.2.2](#) and are disjoint each other. We have the followings:

- $X = (X \setminus Y) \cup (X \cap Y) = A \cup C$ .
- $Y = (Y \setminus X) \cup (X \cap Y) = B \cup C$ .
- $X \cup Y = ((X \setminus Y) \cup (Y \setminus X)) \cup (X \cap Y) = (A \cup B) \cup C$ .

By [Lemma 2.2.9](#),  $X \cup Y$  is finite. □

**Theorem 2.2.11.** *Let  $X_1, X_2, \dots, X_n$  be finite sets for some  $n \in \mathbb{N}$ . Then the union  $\bigcup_{k=1}^n X_k$  is finite, i.e., any finite union of finite sets is finite.*

*Proof.* Let  $P(n)$  be the statement that

$$X_1, X_2, \dots, X_n \text{ are finite sets} \implies \bigcup_{k=1}^n X_k \text{ is finite.}$$

- **Base case:** Let  $X_1$  be a finite set. Then  $\bigcup_{k=1}^1 X_k = X_1$  is finite. Thus  $P(1)$  is true.

- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ .

Let  $X_1, X_2, \dots, X_{n+1}$  be finite sets. Then we have

$$\bigcup_{k=1}^{n+1} X_k = \left( \bigcup_{k=1}^n X_k \right) \cup X_{n+1}.$$

By the inductive hypothesis,  $\bigcup_{k=1}^n X_k$  is finite. Then, by [Theorem 2.2.10](#),

$\bigcup_{k=1}^{n+1} X_k$  is finite. Thus  $P(n+1)$  is true whenever  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ . □

**Lemma 2.2.12.** *Let  $X, Y$  be finite sets. Then the cartesian product  $X \times Y$  is finite.*

*Proof.* Let  $m = |X|$  and  $n = |Y|$ . There exist bijections  $f : J_m \rightarrow X$  and  $g : J_n \rightarrow Y$ . Let  $h : J_{mn} \rightarrow X \times Y$  be a function defined by

$$h(i) = \left( f\left(\left[\frac{i}{n}\right]\right), g\left(i - n\left(\left[\frac{i}{n}\right] - 1\right)\right) \right) \text{ for all } i \in J_{mn}.$$

Then  $h$  is bijective, which implies that  $X \times Y$  is finite.  $\square$

**Theorem 2.2.13.** *Let  $X_1, X_2, \dots, X_n$  be finite sets for some  $n \in \mathbb{N}$ . Then the cartesian product  $\prod_{k=1}^n X_k$  is finite.*

*Proof.* Let  $P(n)$  be the statement that

$$X_1, X_2, \dots, X_n \text{ are finite sets} \implies \prod_{k=1}^n X_k \text{ is finite.}$$

- **Base case:** Let  $X_1$  be a finite set. Then,  $\prod_{k=1}^1 X_k = X_1$  is finite. Thus  $P(1)$  is true.

- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ .

Let  $X_1, X_2, \dots, X_{n+1}$  be finite sets. Then we have

$$\prod_{k=1}^{n+1} X_k = \left( \prod_{k=1}^n X_k \right) \times X_{n+1}.$$

By the inductive hypothesis,  $\prod_{k=1}^n X_k$  is finite. Then, by [Lemma 2.2.12](#),  $\prod_{k=1}^{n+1} X_k$  is finite. Thus  $P(n+1)$  is true whenever  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ .  $\square$

**Theorem 2.2.14.** *Let  $X$  be a nonempty set. Then there exists no surjection  $f : X \rightarrow \mathcal{P}(X)$ , where*

$$\mathcal{P}(X) := \{E \mid E \subseteq X\}$$

*is called the **power set** of  $X$ . This theorem is called **Cantor's theorem**.*

*Proof.* Let  $S$  be a set defined by

$$S := \{x \in X \mid x \notin f(x)\}.$$

Since  $S \subseteq X$ , we have that  $S \in \mathcal{P}(X)$ . Assume that there exists a surjection  $f : X \rightarrow \mathcal{P}(X)$ . Then there exists  $x \in X$  such that  $f(x) = S$ .

If  $x \in S$ , then we have  $x \notin f(x) = S$ , which is a contradiction. If  $x \notin S$ , then we have  $x \in f(x) = S$ , which is also a contradiction. Therefore, there exists no surjection  $f : X \rightarrow \mathcal{P}(X)$ .  $\square$

## 2.3 Countable and Uncountable Sets

A set  $X$  is said to be **countable** if  $\mathbb{N} \sim X$ . A set  $X$  is said to be **uncountable** if it is neither finite nor countable. A set  $X$  is said to be **at most countable** if  $X$  is either finite or countable.

**Theorem 2.3.1.** *Let  $X$  be an infinite set, and let  $Y$  be a finite set. Then  $X \setminus Y$  is infinite.*

*Proof.* Assume that  $X \setminus Y$  is finite. Since  $Y$  is finite, by [Theorem 2.2.10](#),  $(X \setminus Y) \cup Y$  is finite. Then, by [Theorem 2.2.2](#),  $X \subseteq (X \setminus Y) \cup Y$  is finite, which contradicts the assumption. Therefore,  $X \setminus Y$  is infinite.  $\square$

**Theorem 2.3.2.** *Let  $X$  be an infinite set. If there exists an injection  $f : X \rightarrow Y$ , then  $Y$  is infinite.*

*Proof.* Assume that  $Y$  is finite. Let  $g : Y \rightarrow X$  be defined by

$$g(y) = \begin{cases} f^{-1}(y), & \text{if } y \in f(X) \\ f^{-1}(y_0), & \text{Otherwise} \end{cases}$$

where  $y_0 \in f(X)$  is fixed. Then  $g$  is surjective. By [Theorem 2.2.4](#),  $X$  is finite, which is a contradiction. Therefore,  $Y$  is infinite.  $\square$

**Lemma 2.3.3.**  *$\mathbb{N}$  is infinite.*

*Proof.* Assume that  $\mathbb{N}$  is finite. Then there exists a bijection  $f : J_n \rightarrow \mathbb{N}$  for some  $n \in \mathbb{N}$ . Then, by [Theorem 2.2.2](#),  $f(J_n) \subseteq \mathbb{N}$  is finite. By [Theorem 2.2.5](#),  $f(J_n)$  has a greatest element, say  $m$ .

Then we have  $m+1 \in \mathbb{N}$  and  $m < m+1 \notin f(J_n)$ , which contradicts the assumption that  $f$  is surjective. Therefore,  $\mathbb{N}$  is infinite.  $\square$

**Theorem 2.3.4.** *Any countable set is infinite.*

*Proof.* Let  $X$  be a countable set. Then there exists a bijection  $f : \mathbb{N} \rightarrow X$ . By [Theorem 2.3.2](#) and [lemma 2.3.3](#),  $X$  is infinite.  $\square$

*Remark.* Let  $X$  be a countable set. Then there exists a bijection  $f : \mathbb{N} \rightarrow X$ . The elements of  $X$  can be listed as  $X = \{x_1, x_2, x_3, \dots\}$ , where  $x_n = f(n)$  for all  $n \in \mathbb{N}$ . In other words, the elements of  $X$  can be arranged in a sequence.

**Theorem 2.3.5.**  *$\mathbb{Z}$  is countable.*

*Proof.* Let  $f : \mathbb{N} \rightarrow \mathbb{Z}$  be a function defined by

$$f(n) = \begin{cases} -\frac{n}{2}, & \text{if } n \text{ is even} \\ \frac{n-1}{2}, & \text{if } n \text{ is odd} \end{cases}$$

We will show that  $f$  is bijective, which implies that  $\mathbb{Z}$  is countable.

- **Injectivity:** Assume that  $f(n_1) = f(n_2)$  for some  $n_1, n_2 \in \mathbb{N}$ . Since  $f(n) \geq 0$  for odd  $n$  and  $f(n) < 0$  for even  $n$ , we have  $n_1$  and  $n_2$  are either both even or both odd.

- If both  $n_1$  and  $n_2$  are even, then we have  $-\frac{n_1}{2} = -\frac{n_2}{2}$ , which implies  $n_1 = n_2$ .
- If both  $n_1$  and  $n_2$  are odd, then we have  $\frac{n_1-1}{2} = \frac{n_2-1}{2}$ , which implies  $n_1 = n_2$ .

Thus  $f$  is injective.

- **Surjectivity:** Let  $k \in \mathbb{Z}$  be given. We need to find an  $n \in \mathbb{N}$  such that  $f(n) = k$ .

- If  $k \geq 0$ , let  $n = 2k + 1$ . Then we have  $f(n) = \frac{n-1}{2} = \frac{2k}{2} = k$ .
- If  $k < 0$ , let  $n = -2k$ . Then we have  $f(n) = -\frac{n}{2} = -\frac{-2k}{2} = k$ .

Thus  $f$  is surjective.

$\square$

**Lemma 2.3.6.** *Any infinite subset of  $\mathbb{N}$  is countable.*

*Proof.* Let  $X \subseteq \mathbb{N}$  be an infinite subset. Let  $(X_n)_{n=1}^{\infty}$  be a sequence and let  $f : \mathbb{N} \rightarrow X$  be a function defined as follows:

- $X_1 = X, f(1) = \min X_1$
- $X_2 = X_1 \setminus \{f(1)\}, f(2) = \min X_2$
- $\vdots$
- $X_{n+1} = X_n \setminus \{f(n)\}, f(n+1) = \min X_{n+1}$

for each  $n \in \mathbb{N}$ . By [Theorem 2.3.1](#),  $X_n$  is infinite for each  $n \in \mathbb{N}$ , so is nonempty. Then, by [Theorem 1.1.3](#),  $X_n$  has a least element, so  $f$  is well-defined. We will show that  $f$  is bijective, which implies that  $X$  is countable.

- **Injectivity:** Assume that  $n_1 \neq n_2$  for some  $n_1, n_2 \in \mathbb{N}$ . Without loss of generality, assume that  $n_1 < n_2$ . Then we have

$$f(n_1) \in X_{n_1} \text{ and } f(n_2) \in X_{n_2}.$$

Since  $X_{n_2} \subseteq X_{n_1} \setminus \{f(n_1)\}$ , we have  $f(n_2) \neq f(n_1)$ . Thus  $f$  is injective.

- **Surjectivity:** Let  $x \in X$  be given. Let  $S$  be a set defined as

$$S := \{n \in X \mid n < x\}.$$

If  $S = \emptyset$ , then  $f(1) = x$ . Otherwise, by [Lemma 2.2.1](#),  $S \subseteq J_{x-1}$  is finite. Let  $k = |S|$ ,  $S = \{s_1, s_2, \dots, s_k\}$  with  $s_1 < s_2 < \dots < s_k$ . Then we have

- $\min X_1 = \min S \implies X_2 = X \setminus \{s_1\}.$
- $\min X_2 = \min S \setminus \{s_1\} \implies X_3 = X \setminus \{s_1, s_2\}.$
- $\vdots$
- $\min X_k = \min S \setminus \{s_1, s_2, \dots, s_{k-1}\} \implies X_{k+1} = X \setminus S.$

Thus  $f(k+1) = \min X_{k+1} = x$ . Therefore,  $f$  is surjective.

□

**Theorem 2.3.7.** *Any infinite subset of a countable set is countable.*

*Proof.* Let  $X$  be a countable set and let  $Y \subseteq X$  be an infinite subset. Since  $X$  is countable, there exists a bijection  $f : \mathbb{N} \rightarrow X$ . By [Theorem 2.1.5](#),  $f|_{f^{-1}(Y)}^Y : f^{-1}(Y) \rightarrow Y$  is bijective. Then, by [Theorem 2.2.4](#),  $f^{-1}(Y) \subseteq \mathbb{N}$  is infinite.

By [Lemma 2.3.6](#),  $f^{-1}(Y)$  is countable. Then there exists a bijection  $g : \mathbb{N} \rightarrow f^{-1}(Y)$ . The composition  $f|_{f^{-1}(Y)}^Y \circ g : \mathbb{N} \rightarrow Y$  is bijective, which implies that  $Y$  is countable. □

**Theorem 2.3.8.** *Let  $X$  be infinite. If there exists a surjection  $f : \mathbb{N} \rightarrow X$ , then  $X$  is countable.*

*Proof.* For each  $k \in X$ , let  $S_k$  be a set defined as

$$S_k := f^{-1}(\{k\}).$$

Since  $f$  is surjective,  $S_k \neq \emptyset$  and by Theorem 1.1.3,  $S_k$  has a least element for all  $k \in X$ . Let  $g : X \rightarrow \mathbb{N}$  be a function defined by

$$g(k) = \min S_k \text{ for all } k \in X.$$

Assume that  $g(k_1) = g(k_2)$  for some  $k_1, k_2 \in X$ , i.e.,  $\min S_{k_1} = \min S_{k_2}$ . Since  $S_i \cap S_j = \emptyset$  for all  $i, j \in X$  with  $i \neq j$ , we have  $S_{k_1} = S_{k_2} \implies k_1 = k_2$ . Thus  $g$  is injective.

Since  $g$  is injective, by Theorem 2.1.5,  $g|^{g(X)} : X \rightarrow g(X)$  is bijective. Since  $X$  is infinite,  $g(X) \subseteq \mathbb{N}$  is infinite by Theorem 2.3.2. Then, by Lemma 2.3.6,  $g(X)$  is countable, which implies that  $X$  is countable.  $\square$

**Theorem 2.3.9.** *Let  $\{X_n\}_{n \in \mathbb{N}}$  be a sequence of countable sets. Then  $\bigcup_{n=1}^{\infty} X_n$  is countable, i.e., any countable union of countable sets is countable.*

*Proof.* There exists a bijection  $f_n : \mathbb{N} \rightarrow X_n$  for each  $n \in \mathbb{N}$ . For each  $n \in \mathbb{N}$ , we define the followings:

- $t_n := \min T_n$ , where  $T_n := \left\{ k \in \mathbb{N} \mid n \leq \frac{k(k+1)}{2} \right\}$ .
- $s_n := n - \frac{t_n(t_n - 1)}{2} = n - \frac{t_n(t_n + 1)}{2} + t_n \in \mathbb{N}$ .

Since  $2n \in T_n$ ,  $T_n$  is nonempty. Then, by Theorem 1.1.3,  $T_n$  has a least element, so  $t_n$  is well-defined. Since  $\frac{t_n(t_n - 1)}{2} < n \leq \frac{t_n(t_n + 1)}{2}$ , we have

$$t_n - s_n + 1 = \frac{t_n(t_n + 1)}{2} - n + 1 \geq 1,$$

so  $t_n - s_n + 1 \in \mathbb{N}$ . Let  $g : \mathbb{N} \rightarrow \bigcup_{n=1}^{\infty} X_n$  be a function defined by

$$g(k) = f_{s_k}(t_k - s_k + 1) \text{ for all } k \in \mathbb{N}.$$

Let  $x \in \bigcup_{n=1}^{\infty} X_n$  be given. Then there exists some  $m \in \mathbb{N}$  such that  $x \in X_m$ . Since  $f_m$  is surjective, there exists some  $l \in \mathbb{N}$  such that  $f_m(l) = x$ . Let

$$k := \frac{(l+m-1)(l+m-2)}{2} + m \in \mathbb{N}.$$

We will show that  $l+m-2 \notin T_k$  and  $l+m-1 \in T_k$ , which implies that

- $t_k = \min T_k = l+m-1$ .
- $s_k = k - \frac{t_k(t_k-1)}{2} = m$ .
- $g(k) = f_{s_k}(t_k - s_k + 1) = f_m(l+m-1-m+1) = f_m(l) = x$ .

Consider the following inequalities when  $z = l+m-2$  and  $z = l+m-1$ :

- Let  $z = l+m-2$ . Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-2)(l+m-1)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-2)(l+m-1)}{2} \\ &= m > 0. \end{aligned}$$

- Let  $z = l+m-1$ . Then we have

$$\begin{aligned} k - \frac{z(z+1)}{2} &= k - \frac{(l+m-1)(l+m)}{2} \\ &= \frac{(l+m-1)(l+m-2)}{2} + m - \frac{(l+m-1)(l+m)}{2} \\ &= m - \frac{(l+m-1)(l+m) - (l+m-1)(l+m-2)}{2} \\ &= m - (l+m-1) = 1-l \leq 0. \end{aligned}$$

Thus  $g$  is surjective, which implies that  $\bigcup_{n=1}^{\infty} X_n$  is countable by [Theorem 2.3.8](#).  $\square$

**Corollary 2.3.10.** *Let  $X_1, X_2, \dots, X_n$  be countable sets for some  $n \in \mathbb{N}$ . Then*

*$\bigcup_{i=1}^n X_i$  is countable, i.e., any finite union of countable sets is countable.*

*Proof.* Let  $X_{n+1}, X_{n+2}, \dots$  be countable sets. Then, by [Theorem 2.3.9](#),  $\bigcup_{n=1}^{\infty} X_n$  is countable. Assume that  $\bigcup_{i=1}^n X_i$  is finite. Then  $X_1 \subseteq \bigcup_{i=1}^n X_i$  is finite by [Theorem 2.2.2](#), which contradicts [Theorem 2.3.4](#). Thus  $\bigcup_{i=1}^n X_i$  is infinite. Since  $\bigcup_{i=1}^n X_i \subseteq \bigcup_{n=1}^{\infty} X_n$ , by [Theorem 2.3.7](#),  $\bigcup_{i=1}^n X_i$  is countable.  $\square$

**Theorem 2.3.11.** *Let  $X$  be a countable set. If  $Y_x$  is finite for each  $x \in X$ , then the union*

$$Y = \bigcup_{x \in X} Y_x$$

*is countable, i.e., any countable union of finite sets is countable.*

*Proof.* There exists a bijection  $f : \mathbb{N} \rightarrow X$ . Also, for each  $x \in X$ , there exists a bijection  $g_x : J_{n_x} \rightarrow Y_x$  for some  $n_x \in \mathbb{N}$ . Let  $h : \mathbb{N} \rightarrow Y$  be a function defined by

- $h(j) = g_{f(1)}(j)$  for  $j = 1, 2, \dots, n_{f(1)}$ .
- $h\left(\sum_{i=1}^{k-1} n_{f(i)} + m\right) = g_{f(k)}(m)$  for each  $k \in \mathbb{N}$  with  $k \geq 2$ , and  $m = 1, 2, \dots, n_{f(k)}$ .

Then  $h$  is surjective. Therefore, by [Theorem 2.3.8](#),  $Y$  is countable.  $\square$

**Lemma 2.3.12.**  $\mathbb{N} \times \mathbb{N}$  is countable.

*Proof.* We will prove this lemma by explicitly constructing a bijection from  $\mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ . Let  $f : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$  be a function defined by

$$f(m, n) = \frac{(m+n-1)(m+n-2)}{2} + m$$

for each  $(m, n) \in \mathbb{N} \times \mathbb{N}$ . We will show that  $f$  is bijective.

- **Injectivity:** Assume that  $(m_1, n_1) \neq (m_2, n_2)$ . If  $m_1 + n_1 = m_2 + n_2$ , then  $m_1 \neq m_2$ , so  $f(m_1, n_1) \neq f(m_2, n_2)$ . Otherwise,  $m_1 + n_1 \neq m_2 + n_2$ .

Without loss of generality, assume  $m_1 + n_1 < m_2 + n_2$ . Then, since  $m_1 +$

$n_1 + 1 \leq m_2 + n_2$ , we have

$$\begin{aligned} f(m_1, n_1) &= \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + m_1 \\ &\leq \frac{(m_1 + n_1 - 1)(m_1 + n_1 - 2)}{2} + (m_1 + n_1 - 1) \\ &= \frac{(m_1 + n_1)(m_1 + n_1 - 1)}{2} \\ &\leq \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} \\ &< \frac{(m_2 + n_2 - 1)(m_2 + n_2 - 2)}{2} + m_2 = f(m_2, n_2), \end{aligned}$$

Thus  $f(m_1, n_1) \neq f(m_2, n_2)$ , so  $f$  is injective.

- **Surjectivity:** Let  $k \in \mathbb{N}$  be given. Let  $t_k, s_k$  be defined as in the proof of [Theorem 2.3.9](#). Then we have  $f(s_k, t_k - s_k + 1) = k$ . Thus  $f$  is surjective.

□

**Theorem 2.3.13.** *Let  $X$  and  $Y$  be countable sets. Then the cartesian product  $X \times Y$  is countable.*

*Proof.* There exist bijections  $f : \mathbb{N} \rightarrow X$  and  $g : \mathbb{N} \rightarrow Y$ . Let  $h : \mathbb{N} \times \mathbb{N} \rightarrow X \times Y$  be a function defined by

$$h(m, n) = (f(m), g(n)) \text{ for all } (m, n) \in \mathbb{N} \times \mathbb{N}.$$

Then  $h$  is bijective. By [Lemma 2.3.12](#),  $X \times Y$  is countable.

□

**Theorem 2.3.14.** *Let  $X_1, X_2, \dots, X_n$  be countable sets for some  $n \in \mathbb{N}$ . Then  $\prod_{i=1}^n X_i$  is countable, i.e., any finite product of countable sets is countable.*

*Proof.* Let  $P(n)$  be the statement that

$$X_1, X_2, \dots, X_n \text{ are countable sets} \implies \prod_{i=1}^n X_i \text{ is countable.}$$

- **Base case:** Let  $X_1$  be a countable set. Then  $\prod_{i=1}^1 X_i = X_1$  is countable. Thus  $P(1)$  is true.

- **Inductive step:** Assume that  $P(n)$  is true for some  $n \in \mathbb{N}$ .

Let  $X_1, X_2, \dots, X_{n+1}$  be countable sets. Then we have

$$\prod_{i=1}^{n+1} X_i = \left( \prod_{i=1}^n X_i \right) \times X_{n+1}$$

By the inductive hypothesis,  $\prod_{i=1}^n X_i$  is countable. Then, by [Theorem 2.3.13](#),

$\prod_{i=1}^{n+1} X_i$  is countable. Thus  $P(n+1)$  is true when  $P(n)$  is true.

By the principle of mathematical induction,  $P(n)$  is true for all  $n \in \mathbb{N}$ .  $\square$

**Theorem 2.3.15.**  $\mathbb{Q}$  is countable.

*Proof.* Any  $q \in \mathbb{Q}$  can be expressed as  $q = \frac{m}{n}$  for some  $m \in \mathbb{Z}$  and  $n \in \mathbb{N}$ . Let  $f : \mathbb{Z} \times \mathbb{N} \rightarrow \mathbb{Q}$  be a function defined by

$$f(m, n) = \frac{m}{n} \text{ for all } (m, n) \in \mathbb{Z} \times \mathbb{N}.$$

Then  $f$  is surjective. By [Theorems 2.3.5](#) and [2.3.13](#),  $\mathbb{Z} \times \mathbb{N}$  is countable. Then there exists a bijection  $g : \mathbb{N} \rightarrow \mathbb{Z} \times \mathbb{N}$ .

By [Theorem 2.1.7](#),  $f \circ g : \mathbb{N} \rightarrow \mathbb{Q}$  is surjective. Since  $\mathbb{N} \subseteq \mathbb{Q}$ , by [Lemma 2.3.3](#) and [theorem 2.2.2](#),  $\mathbb{Q}$  is infinite. Therefore, by [Theorem 2.3.8](#),  $\mathbb{Q}$  is countable.  $\square$

**Lemma 2.3.16.**  $X := \prod_{n=1}^{\infty} \{0, 1\}$  is uncountable.

*Proof.* For each  $n \in \mathbb{N}$ , let  $e_n \in X$  be a sequence defined by

$$e_{nm} = \begin{cases} 1 & \text{if } m = n \\ 0 & \text{Otherwise} \end{cases}$$

Let  $f : \mathbb{N} \rightarrow X$  be a function defined by

$$f(n) = e_n \text{ for all } n \in \mathbb{N}.$$

Then  $f$  is injective. By [Lemma 2.3.3](#) and [theorem 2.3.2](#),  $X$  is infinite.

Assume that  $X$  is countable. Then, there exists a bijection  $g : \mathbb{N} \rightarrow X$ . Let  $X = \{g(1), g(2), g(3), \dots\}$ , where

- $g(1) = (a_{11}, a_{12}, a_{13}, \dots)$ .
- $g(2) = (a_{21}, a_{22}, a_{23}, \dots)$ .
- $\vdots$

Let  $x = (b_1, b_2, b_3, \dots) \in X$  be a sequence defined by

$$b_n = 1 - a_{nn} \text{ for all } n \in \mathbb{N}.$$

Then,  $x \neq g(n)$  for all  $n \in \mathbb{N}$ , which contradicts the assumption that  $g$  is surjective. Therefore,  $X$  is uncountable.  $\square$

**Theorem 2.3.17.**  $\mathbb{R}$  is uncountable.

*Proof.* Since  $\mathbb{N} \subseteq \mathbb{R}$ , by Lemma 2.3.3 and theorem 2.2.2,  $\mathbb{R}$  is infinite. Let  $f : \mathbb{N} \rightarrow [0, 1)$  be a function defined by

$$f(n) = \frac{1}{n+1} \text{ for all } n \in \mathbb{N}.$$

Then  $f$  is injective. By Lemma 2.3.3 and theorem 2.3.2,  $[0, 1)$  is infinite. We will prove that  $[0, 1) \subseteq \mathbb{R}$  is uncountable, which implies that  $\mathbb{R}$  is uncountable by Theorem 2.3.7.

Let  $g : [0, 1) \rightarrow \prod_{n=1}^{\infty} \{0, 1\}$  be a function defined by

$$g(x) = (a_n^x)_{n=1}^{\infty} \text{ for all } x \in [0, 1)$$

where  $(a_n^x)_{n=1}^{\infty}$  is the greedy base 2 expansion of  $x$ . Let  $Y$  be a set defined by

$$Y := g([0, 1)) \subseteq \prod_{n=1}^{\infty} \{0, 1\}.$$

Then, by Theorems 1.5.9 and 2.1.5,  $g|_Y$  is bijective.

Assume that  $[0, 1)$  is countable. Then there exists a bijection  $h : \mathbb{N} \rightarrow [0, 1)$ . By Theorem 2.1.7,  $g|_Y \circ h : \mathbb{N} \rightarrow Y$  is bijective, so  $Y$  is countable. Let  $Z :=$

$\prod_{n=1}^{\infty} \{0, 1\} \setminus Y$ . By Theorems 1.5.10 and 1.5.11, we have

$$Z = \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid a_n = 1 \text{ for all } n \geq N \text{ for some } N \in \mathbb{N} \right\}.$$

Let  $Z_N$  be a set defined by

$$Z_N := \left\{ (a_n)_{n=1}^{\infty} \in \prod_{n=1}^{\infty} \{0, 1\} \mid a_n = 1 \text{ for all } n \geq N \right\} \text{ for each } N \in \mathbb{N}.$$

Let  $F_N : \prod_{n=1}^{N-1} \{0, 1\} \rightarrow Z_N$  be a function defined by

$$F_N\left((a_n)_{n=1}^{N-1}\right) = (a_n)_{n=1}^{N-1} \cup (1)_{n=N}^{\infty} \text{ for all } (a_n)_{n=1}^{N-1} \in \prod_{n=1}^{N-1} \{0, 1\}.$$

Then  $F_N$  is surjective. By [Theorems 2.2.4](#) and [2.2.13](#),  $Z_N$  is finite. Then, by [Theorem 2.3.11](#),  $Z = \bigcup_{N=1}^{\infty} Z_N$  is countable.

By [Corollary 2.3.10](#),  $\prod_{n=1}^{\infty} \{0, 1\} = Y \cup Z$  is countable. [Lemma 2.3.16](#) shows that this is a contradiction. Therefore,  $[0, 1)$  is uncountable.  $\square$

## 2.4 Metric Spaces

Let  $X$  be a set. A function  $d : X \times X \rightarrow \mathbb{R}$  is called a **metric** on  $X$  if for any  $p, q \in X$ , we have

$$\text{(A)} \quad d(p, q) \geq 0 \text{ and } d(p, q) = 0 \iff p = q.$$

$$\text{(B)} \quad d(p, q) = d(q, p).$$

$$\text{(C)} \quad d(p, r) \leq d(p, q) + d(q, r) \text{ for any } r \in X.$$

(C) is called the **triangle inequality**. The pair  $(X, d)$  is called a **metric space**.

**Theorem 2.4.1.** For some  $n \in \mathbb{N}$ , let  $d : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$  be a function defined by

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{\sum_{k=1}^n (x_k - y_k)^2}$$

for all  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ . Then  $d$  is a metric on  $\mathbb{R}^n$ . Especially,  $d$  is called the **standard metric** on  $\mathbb{R}^n$ .

*Proof.* Let  $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$  be given.

- $d(\mathbf{x}, \mathbf{y}) \geq 0$  is trivial. Also, we have

$$\begin{aligned} d(\mathbf{x}, \mathbf{y}) = 0 &\iff \|\mathbf{x} - \mathbf{y}\| = 0 \\ &\iff x_k = y_k \text{ for all } k = 1, 2, \dots, n \\ &\iff \mathbf{x} = \mathbf{y}. \end{aligned}$$

- $d(\mathbf{x}, \mathbf{y}) = d(\mathbf{y}, \mathbf{x})$  is trivial.
- By [Theorem 1.8.2](#), we have

$$\begin{aligned} d(\mathbf{x}, \mathbf{z}) &= \|\mathbf{x} - \mathbf{z}\| = \|(\mathbf{x} - \mathbf{y}) + (\mathbf{y} - \mathbf{z})\| \\ &\leq \|\mathbf{x} - \mathbf{y}\| + \|\mathbf{y} - \mathbf{z}\| = d(\mathbf{x}, \mathbf{y}) + d(\mathbf{y}, \mathbf{z}) \end{aligned}$$

Therefore,  $d$  is a metric on  $\mathbb{R}^n$ . □

**Definition 2.4.2.** Let  $k \in \mathbb{N}$  be given. A set  $E \subseteq \mathbb{R}^k$  defined by

$$E := \left\{ \mathbf{x} = (x_1, x_2, \dots, x_k) \in \mathbb{R}^k \mid a_j \leq x_j \leq b_j \text{ for all } j = 1, 2, \dots, k \right\}$$

is called an  **$k$ -cell**, where  $-\infty < a_j \leq b_j < +\infty$  for all  $j = 1, 2, \dots, k$ .

**Definition 2.4.3.** Let  $(X, d)$  be a metric space. For any  $p \in X$  and  $r \in \mathbb{R}^+$ , we define the **neighborhood** of  $p$  with radius  $r$  as

$$N_r(p) := \{q \in X \mid d(p, q) < r\}.$$

Also, we define the **deleted neighborhood** of  $p$  with radius  $r$  as

$$\tilde{N}_r(p) := N_r(p) \setminus \{p\} = \{q \in X \mid 0 < d(p, q) < r\}.$$

Especially, if  $X = \mathbb{R}^n$  and  $d$  is the standard metric on  $\mathbb{R}^n$  for some  $n \in \mathbb{N}$ , then  $N_r(p)$  is called the **open ball** of radius  $r$  centered at  $p$ , which is denoted by  $B_r(p)$ .

**Definition 2.4.4.** Let  $(X, d)$  be a metric space and  $E \subseteq X$ . Let  $p \in X$  be given.

- $p$  is called an **limit point** of  $E$  if for any  $r \in \mathbb{R}^+$ , we have  $E \cap \tilde{N}_r(p) \neq \emptyset$ .
- $p$  is called a **isolated point** of  $E$  if there exists some  $r \in \mathbb{R}^+$  such that  $N_r(p) \cap E = \{p\}$ .
- $p$  is called an **interior point** of  $E$  if there exists some  $r \in \mathbb{R}^+$  such that  $N_r(p) \subseteq E$ .

**Definition 2.4.5.** Let  $(X, d)$  be a metric space and  $E \subseteq X$ .

- $E$  is said to be **open** if any point of  $E$  is an interior point of  $E$ .
- $E$  is said to be **closed** if any limit point of  $E$  belongs to  $E$ .

**Definition 2.4.6.** Let  $(X, d)$  be a metric space and  $E \subseteq X$ .

- $E$  is said to be **bounded** if there exists some  $p \in X$  and  $r \in \mathbb{R}^+$  such that  $E \subseteq N_r(p)$ .
- $E$  is said to be **perfect** if any point of  $E$  is a limit point of  $E$ .
- $E$  is said to be **dense** if any point of  $X$  is either a limit point of  $E$  or a point of  $E$ .

**Theorem 2.4.7.** Let  $(X, d)$  be a metric space. For any  $p \in X$  and  $r \in \mathbb{R}^+$ , the neighborhood  $N_r(p)$  is open.

*Proof.* Let  $q \in N_r(p)$  be given. Then we have  $d(p, q) < r \implies \varepsilon := r - d(p, q) > 0$ . Let  $s \in N_\varepsilon(q)$  be given. Then we have

$$d(q, s) < \varepsilon \implies d(p, s) \leq d(p, q) + d(q, s) < d(p, q) + \varepsilon = r.$$

Thus  $s \in N_r(p)$ , which implies that  $N_\varepsilon(q) \subseteq N_r(p)$ . Since  $q$  is arbitrary,  $N_r(p)$  is open.  $\square$

**Theorem 2.4.8.** Let  $(X, d)$  be a metric space and  $E \subseteq X$ . Let  $p$  be a limit point of  $E$ . Then  $E \cap N_r(p)$  is infinite for all  $r \in \mathbb{R}^+$ .

*Proof.* Assume that there exists some  $r \in \mathbb{R}^+$  such that  $E \cap N_r(p)$  is finite. Let  $S$  be a set defined by

$$S := (E \cap N_r(p)) \setminus \{p\} = E \cap \tilde{N}_r(p).$$

Since  $p$  is a limit point of  $E$ , we have  $S \neq \emptyset$ . Since  $E \cap N_r(p)$  is finite,  $S \subseteq E \cap N_r(p)$  is also finite by [Theorem 2.2.2](#). Let  $T$  be a set defined by

$$T := \{d(p, q) \mid q \in S\} \neq \emptyset.$$

Since  $p \notin S$ , we have  $0 \notin T$ . Let  $f : S \rightarrow T$  be a function defined by

$$f(q) = d(p, q).$$

Then  $f$  is surjective. By [Theorem 2.2.4](#),  $T$  is finite. Then, by [Theorem 2.2.5](#),  $T$  has a minimum element, say  $\varepsilon = \min T > 0$ . Then we have

- $N_\varepsilon(p) \subset N_r(p)$ .
- $E \cap \tilde{N}_\varepsilon(p) = \emptyset$ .

This contradicts the assumption that  $p$  is a limit point of  $E$ . Therefore,  $E \cap N_r(p)$  is infinite.  $\square$

**Theorem 2.4.9.** *Let  $(X, d)$  be a metric space and  $E \subseteq X$ . Then we have*

$$E \text{ is open} \iff E^c \text{ is closed.}$$

*Proof.* (  $\implies$  ) Assume that  $E$  is open. Let  $p$  be a limit point of  $E^c$ . Then, for any  $r \in \mathbb{R}^+$ , we have

$$E^c \cap \tilde{N}_r(p) \neq \emptyset.$$

If  $p \in E$ , then there exists some  $\varepsilon \in \mathbb{R}^+$  such that  $N_\varepsilon(p) \subseteq E$ . Then we have

$$E^c \cap \tilde{N}_\varepsilon(p) = \emptyset,$$

which is a contradiction. Therefore, we have  $p \in E^c$ , which implies that  $E^c$  is closed.

(  $\impliedby$  ) Assume that  $E^c$  is closed. Let  $p$  be a point of  $E$ . If  $p$  is not an interior point of  $E$ , then for any  $r \in \mathbb{R}^+$ , we have

$$E^c \cap N_r(p) \neq \emptyset.$$

Since  $p \in E \implies p \notin E^c$ , we have

$$E^c \cap \tilde{N}_r(p) \neq \emptyset,$$

which implies that  $p$  is a limit point of  $E^c$ . Since  $E^c$  is closed, we have  $p \in E^c$ , which is a contradiction. Therefore,  $p$  is an interior point of  $E$ . Since  $p$  is arbitrary,  $E$  is open.  $\square$

**Corollary 2.4.10.** *Let  $(X, d)$  be a metric space and  $E \subseteq X$ . Then we have*

$$E \text{ is closed} \iff E^c \text{ is open.}$$

**Theorem 2.4.11.** *Let  $(X, d)$  be a metric space. Let  $E_y \subseteq X$  be an open set for each  $y \in Y$ . Then we have*

$$\bigcup_{y \in Y} E_y \text{ is open}$$

*whether  $Y$  is finite, countable, or uncountable. This theorem states that an arbitrary union of open sets is open.*

*Proof.* Let  $p \in \bigcup_{y \in Y} E_y$  be given. Then there exists some  $y_0 \in Y$  such that  $p \in E_{y_0}$ .

Since  $E_{y_0}$  is open, there exists some  $r \in \mathbb{R}^+$  such that  $N_r(p) \subseteq E_{y_0}$ . Then we have

$$N_r(p) \subseteq E_{y_0} \subseteq \bigcup_{y \in Y} E_y,$$

which implies that  $p$  is an interior point of  $\bigcup_{y \in Y} E_y$ . Since  $p$  is arbitrary,  $\bigcup_{y \in Y} E_y$  is open.  $\square$

**Corollary 2.4.12.** *Let  $(X, d)$  be a metric space. Let  $E_y \subseteq X$  be a closed set for each  $y \in Y$ . Then we have*

$$\bigcap_{y \in Y} E_y \text{ is closed}$$

*whether  $Y$  is finite, countable, or uncountable. This corollary states that an arbitrary intersection of closed sets is closed.*

*Proof.* By Corollary 2.4.10,  $(E_y)^c$  is open for each  $y \in Y$ . Then, by Theorem 2.4.11, we have

$$\bigcup_{y \in Y} (E_y)^c \text{ is open.}$$

By Theorem 2.4.9, we have

$$\bigcap_{y \in Y} E_y = \left( \bigcup_{y \in Y} (E_y)^c \right)^c \text{ is closed.}$$

$\square$

**Theorem 2.4.13.** *Let  $(X, d)$  be a metric space. Let  $E_1, E_2, \dots, E_n \subseteq X$  be open sets for some  $n \in \mathbb{N}$ . Then we have*

$$\bigcap_{k=1}^n E_k \text{ is open.}$$

*This theorem states that a finite intersection of open sets is open.*

*Proof.* Let  $p \in \bigcap_{k=1}^n E_k$  be given. Then we have  $p \in E_j$  for all  $j = 1, 2, \dots, n$ .

Since  $E_j$  is open, there exists some  $r_j \in \mathbb{R}^+$  such that  $N_{r_j}(p) \subseteq E_j$  for each  $j = 1, 2, \dots, n$ . Let  $R$  be a set defined by

$$R := \{r_j \mid j = 1, 2, \dots, n\} \subseteq \mathbb{R}^+.$$

Let  $f : J_n \rightarrow R$  be a function defined by

$$f(j) = r_j.$$

Then  $f$  is surjective. By [Lemma 2.2.1](#) and [theorem 2.2.4](#),  $R$  is finite. Then, by [Theorem 2.2.5](#),  $R$  has a minimum element, say  $r = \min R > 0$ . Then we have

$$N_r(p) \subseteq N_{r_j}(p) \subseteq E_j \text{ for all } j = 1, 2, \dots, n \implies N_r(p) \subseteq \bigcap_{k=1}^n E_k,$$

which implies that  $p$  is an interior point of  $\bigcap_{k=1}^n E_k$ . Since  $p$  is arbitrary,  $\bigcap_{k=1}^n E_k$  is open.  $\square$

**Corollary 2.4.14.** *Let  $(X, d)$  be a metric space. Let  $E_1, E_2, \dots, E_n \subseteq X$  be closed sets for some  $n \in \mathbb{N}$ . Then we have*

$$\bigcup_{k=1}^n E_k \text{ is closed.}$$

*This corollary states that a finite union of closed sets is closed.*

*Proof.* By [Corollary 2.4.10](#),  $(E_k)^c$  is open for each  $k = 1, 2, \dots, n$ . Then, by [Theorem 2.4.13](#), we have

$$\bigcap_{k=1}^n (E_k)^c \text{ is open.}$$

By [Theorem 2.4.9](#), we have

$$\bigcup_{k=1}^n E_k = \left( \bigcap_{k=1}^n (E_k)^c \right)^c \text{ is closed.}$$

$\square$

**Definition 2.4.15.** Let  $(X, d)$  be a metric space and  $E \subseteq X$ . Let  $E'$  be the set of all limit points of  $E$ . The **closure** of  $E$ , which is denoted by  $\overline{E}$ , is defined by

$$\overline{E} := E \cup E'.$$

**Theorem 2.4.16.** *Let  $(X, d)$  be a metric space and  $E \subseteq X$ . Then we have the following properties:*

(A)  $\overline{E}$  is closed.

(B)  $E = \overline{E} \iff E$  is closed.

(C) For any closed subset  $F \subseteq X$  such that  $E \subseteq F$ , we have  $\overline{E} \subseteq F$ .

(C) states that the closure of  $E$  is the smallest closed set containing  $E$ .

*Proof.* (A) We will show that  $(\overline{E})^c$  is open. Let  $p \in (\overline{E})^c$  be given. Since  $p \notin E$  and  $p \notin E'$ , there exists some  $r \in \mathbb{R}^+$  such that

$$E \cap \tilde{N}_r(p) = \emptyset \implies E \cap N_r(p) = \emptyset \implies N_r(p) \subseteq E^c.$$

By Theorem 2.4.7,  $N_r(p)$  is open. Then, for any  $q \in N_r(p)$ , there exists  $\varepsilon \in \mathbb{R}^+$  such that

$$\begin{aligned} N_\varepsilon(q) \subseteq N_r(p) \subseteq E^c &\implies E \cap N_\varepsilon(q) = \emptyset \\ &\implies E \cap (\tilde{N}_\varepsilon(q) \cup \{q\}) = \emptyset \\ &\implies (E \cap \tilde{N}_\varepsilon(q)) \cup (E \cap \{q\}) = \emptyset \\ &\implies E \cap \tilde{N}_\varepsilon(q) = \emptyset \\ &\implies q \notin E'. \end{aligned}$$

Since  $q$  is arbitrary, we have

$$N_r(p) \subseteq (E')^c \implies N_r(p) \subseteq (\overline{E})^c = E^c \cap (E')^c,$$

which implies that  $p$  is an interior point of  $(\overline{E})^c$ . Since  $p$  is arbitrary,  $(\overline{E})^c$  is open, which implies that  $\overline{E}$  is closed by Corollary 2.4.10.

(B)  $E$  is closed  $\iff E' \subseteq E \iff E \cup E' = E \iff \overline{E} = E$ .

(C) Let  $F \subseteq X$  be a closed set such that  $E \subseteq F$ . Let  $p \in E'$  be given. Then, for any  $r \in \mathbb{R}^+$ , we have

$$E \cap \tilde{N}_r(p) \neq \emptyset \implies F \cap \tilde{N}_r(p) \neq \emptyset,$$

which implies that  $p \in F' \subseteq F$ . Therefore, we have  $E' \subseteq F$ . Since  $E \subseteq F$ , we have  $\overline{E} = E \cup E' \subseteq F$ . □

**Theorem 2.4.17.** Let  $(\mathbb{R}, \|\cdot\|)$  be given. Let  $E \subseteq \mathbb{R}$  be nonempty and bounded above. Then we have

$$\sup E \in \overline{E},$$

which implies that

$$E \text{ is closed} \implies \sup E \in E.$$

*Proof.* By the *least upper bound property*, there exists  $\alpha = \sup E \in \mathbb{R}$ . Assume that  $\alpha \in E$ . Then we have

$$E \subseteq \overline{E} \implies \alpha \in \overline{E}.$$

Otherwise, for any  $x \in \mathbb{R}^+$ ,  $\alpha - x$  is not an upper bound of  $E$ . Thus there exists some  $p \in E$  such that

$$\alpha - x < p < \alpha \implies p \in \tilde{N}_x(\alpha) \cap E,$$

which implies that  $\alpha \in E'$ . Therefore, we have

$$E' \subseteq \overline{E} \implies \alpha \in \overline{E}.$$

□

**Theorem 2.4.18.** Let  $(X, d)$  be a metric space and  $E \subseteq X$ . Then we have

$$E \setminus E' = \{p \in X \mid p \text{ is an isolated point of } E\}.$$

*Proof.* ( $\subseteq$ ) Let  $p \in E \setminus E'$  be given. Since  $p \notin E'$ , there exists some  $r \in \mathbb{R}^+$  such that

$$E \cap \tilde{N}_r(p) = \emptyset \implies (E \cap N_r(p)) \setminus \{p\} = \emptyset.$$

Since  $p \in E$  and  $p \in N_r(p)$ , we have

$$E \cap N_r(p) = \{p\},$$

which implies that  $p$  is an isolated point of  $E$ .

( $\supseteq$ ) Let  $p$  be an isolated point of  $E$ . Then there exists some  $r \in \mathbb{R}^+$  such that

$$E \cap N_r(p) = \{p\}.$$

Then we have

- $E \cap \tilde{N}_r(p) = \emptyset \implies p \notin E'$ .
- $p \in E$ .

Therefore, we have  $p \in E \setminus E'$ .

□

**Definition 2.4.19.** Let  $(X, d)$  be a metric space and  $E \subseteq X$ . The **boundary** of  $E$ , which is denoted by  $\partial E$ , is defined by

$$\partial E := \overline{E} \cap \overline{E^c}.$$

**Lemma 2.4.20.** *Let  $(\mathbb{R}^n, \|\cdot\|)$  be given for some  $n \in \mathbb{N}$ . Then we have*

$$\tilde{N}_r(\mathbf{p}) \neq \emptyset \quad \text{for all } \mathbf{p} \in \mathbb{R}^n \text{ and } r \in \mathbb{R}^+.$$

*Proof.* Let  $\mathbf{p} = (p_1, p_2, \dots, p_n) \in \mathbb{R}^n$  and  $r \in \mathbb{R}^+$  be given. Then we have

$$d\left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1, \mathbf{p}\right) = \left\|\left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1\right) - \mathbf{p}\right\| = \left\|\frac{r}{2}\mathbf{e}_1\right\| = \frac{r}{2}\|\mathbf{e}_1\| = \frac{r}{2} < r.$$

where  $\mathbf{e}_1 = (1, 0, 0, \dots, 0) \in \mathbb{R}^n$ . Therefore, we have

$$\mathbf{p} \neq \left(\mathbf{p} + \frac{r}{2}\mathbf{e}_1\right) \in \tilde{N}_r(\mathbf{p}) \implies \tilde{N}_r(\mathbf{p}) \neq \emptyset.$$

□

**Theorem 2.4.21.** *Let  $(\mathbb{R}^n, \|\cdot\|)$  be given for some  $n \in \mathbb{N}$  and  $E \subseteq \mathbb{R}^n$ . Then we have*

$$\partial E = (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)'),$$

*which states that the boundary of  $E$  consists the following three disjoint sets:*

- *The set of all isolated points of  $E$ .*
- *The set of all isolated points of  $E^c$ .*
- *The set of all limit points of both  $E$  and  $E^c$ .*

*Proof.* By the definition of the boundary, we have

$$\begin{aligned} \partial E &= \overline{E} \cap \overline{E^c} \\ &= (E \cup E') \cap (E^c \cup (E^c)') \\ &= (E \cap E^c) \cup (E \cap (E^c)') \cup (E' \cap E^c) \cup (E' \cap (E^c)'). \end{aligned}$$

Consider the following equivalences:

- $E = (E \setminus E') \cup (E \cap E')$ .
- $E^c = (E^c \setminus (E^c)') \cup (E^c \cap (E^c)').$

Then we have the following equalities:

- $E \cap (E^c)' = ((E \setminus E') \cap (E^c)') \cup ((E \cap E') \cap (E^c)').$
- $E' \cap E^c = (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c \cap (E^c)').$

Consider the last two terms of the above equalities:

- $((E \cap E') \cap (E^c)') \subseteq E' \cap (E^c)'.$
- $(E' \cap (E^c \cap (E^c)')) \subseteq E' \cap (E^c)'.$

Then we have

$$\begin{aligned}\partial E &= \emptyset \cup ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)') \\ &= ((E \setminus E') \cap (E^c)') \cup (E' \cap (E^c \setminus (E^c)')) \cup (E' \cap (E^c)')\end{aligned}$$

For any  $\mathbf{p} \in E \setminus E'$ , there exists some  $r \in \mathbb{R}^+$  such that

$$E \cap \tilde{N}_r(\mathbf{p}) = \emptyset \implies \tilde{N}_r(\mathbf{p}) \subseteq E^c.$$

By Lemma 2.4.20, we have

$$\tilde{N}_s(\mathbf{p}) \cap \tilde{N}_r(\mathbf{p}) = \tilde{N}_{\min(s,r)}(\mathbf{p}) \neq \emptyset \quad \text{for all } s \in \mathbb{R}^+.$$

Then, for any  $s \in \mathbb{R}^+$ , we have

$$\begin{aligned}\tilde{N}_s(\mathbf{p}) \cap \tilde{N}_r(\mathbf{p}) \neq \emptyset &\implies \tilde{N}_s(\mathbf{p}) \cap (\tilde{N}_r(\mathbf{p}) \cap E^c) \neq \emptyset \\ &\implies \emptyset \neq (\tilde{N}_s(\mathbf{p}) \cap E^c) \cap \tilde{N}_r(\mathbf{p}) \subseteq \tilde{N}_s(\mathbf{p}) \cap E^c \\ &\implies E^c \cap \tilde{N}_s(\mathbf{p}) \neq \emptyset,\end{aligned}$$

which implies that  $\mathbf{p} \in (E^c)'$ . Then we have

$$E \setminus E' \subseteq (E^c)' \implies (E \setminus E') \cap (E^c)' = E \setminus E'.$$

Similarly, we can show that

$$E^c \setminus (E^c)' \subseteq E' \implies E' \cap (E^c \setminus (E^c)') = E^c \setminus (E^c)'.$$

Therefore, we have

$$\partial E = (E \setminus E') \cup (E^c \setminus (E^c)') \cup (E' \cap (E^c)').$$

□

*Remark.* Let  $(X, d)$  be a metric space and  $Y \subseteq X$ . Then,  $(Y, d|_{Y \times Y})$  is also a metric space, since

$$d|_{Y \times Y} : Y \times Y \rightarrow \mathbb{R}$$

satisfies the properties of a metric. We call  $(Y, d|_{Y \times Y})$  a **subspace** of  $(X, d)$ . For the sake of simplicity, we will denote  $d|_{Y \times Y}$  by  $d$  and write  $(Y, d)$  instead of  $(Y, d|_{Y \times Y})$ .

**Definition 2.4.22.** Let  $(X, d)$  be a metric space and  $E \subseteq Y \subseteq X$ .  $E$  is called **open relative** to  $Y$  if  $E$  is open in the subspace  $(Y, d)$ . That is, for any  $p \in E$ , there exists some  $r \in \mathbb{R}^+$  such that

$$Y \cap N_r(p) \subseteq E.$$

**Theorem 2.4.23.** *Let  $(X, d)$  be a metric space and  $E \subseteq Y \subseteq X$ . Then  $E$  is open relative to  $Y$  if and only if there exists an open set  $V$  in  $X$  such that  $E = V \cap Y$ .*

*Proof.* ( $\implies$ ) Let  $E$  be an open set relative to  $Y$ . Then, for each  $p \in E$ , there exists some  $r_p \in \mathbb{R}^+$  such that

$$Y \cap N_{r_p}(p) \subseteq E.$$

Let  $V$  be a set defined by

$$V := \bigcup_{p \in E} N_{r_p}(p).$$

By Theorems 2.4.7 and 2.4.11,  $V$  is open in  $X$ . Since  $p \in N_{r_p}(p) \subseteq V$  for each  $p \in E$ , we have

- $E \subseteq Y$  and  $E \subseteq V \implies E \subseteq V \cap Y$ .
- $\bigcup_{p \in E} (Y \cap N_{r_p}(p)) \subseteq E \implies Y \cap \bigcup_{p \in E} N_{r_p}(p) \subseteq E \implies Y \cap V \subseteq E$ .

Therefore, we have  $E = V \cap Y$ .

( $\impliedby$ ) Let  $V$  be an open set in  $X$  such that  $E = V \cap Y$ . Then, for each  $p \in E \subseteq V$ , there exists some  $r_p \in \mathbb{R}^+$  such that

$$N_{r_p}(p) \subseteq V \implies Y \cap N_{r_p}(p) \subseteq V \cap Y = E.$$

□

## 2.5 Compact Sets

Let  $(X, d)$  be a metric space and  $E \subseteq X$ . Let  $A$  be a set and let  $G_\alpha$  be an open set in  $X$  for each  $\alpha \in A$ . Then the collection  $\{G_\alpha\}_{\alpha \in A}$  is called an **open cover** of  $E$  if

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

A collection  $\{G_\alpha\}_{\alpha \in S}$  is called a **subcover** of  $E$  if

- $S \subseteq A$ .
- $E \subseteq \bigcup_{\alpha \in S} G_\alpha$ .

$E$  is called **compact** if any open cover of  $E$  has a finite subcover. That is, for each open cover  $\{G_\alpha\}_{\alpha \in A}$  of  $E$ , there exists a subset  $S_A \subseteq A$  such that

- $S_A$  is finite.
- $E \subseteq \bigcup_{\alpha \in S_A} G_\alpha$ .

**Theorem 2.5.1.** *Let  $(X, d)$  be a metric space and  $E \subseteq Y \subseteq X$ . Then  $E$  is compact relative to  $Y$  if and only if  $E$  is compact relative to  $X$ .*

*Proof.* ( $\implies$ ) Let  $E$  be a compact set relative to  $Y$ . Let  $\{G_\alpha\}_{\alpha \in A}$  be an open cover of  $E$  relative to  $X$ . Then we have

$$E \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Let  $H_\alpha = G_\alpha \cap Y$  for each  $\alpha \in A$ . By Theorem 2.4.23,  $H_\alpha$  is open relative to  $Y$ . Since  $E \subseteq Y$ , we have

$$E \subseteq \left( \bigcup_{\alpha \in A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in A} (G_\alpha \cap Y) = \bigcup_{\alpha \in A} H_\alpha.$$

Thus  $\{H_\alpha\}_{\alpha \in A}$  is an open cover of  $E$  relative to  $Y$ . Since  $E$  is compact relative to  $Y$ , there exists a finite subcover  $\{H_\alpha\}_{\alpha \in S_A}$  of  $E$  relative to  $Y$ . Then we have

$$E \subseteq \bigcup_{\alpha \in S_A} H_\alpha = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in S_A} G_\alpha.$$

Thus  $\{G_\alpha\}_{\alpha \in S_A}$  is a finite subcover of  $E$  relative to  $X$ . Therefore,  $E$  is compact relative to  $X$ .

( $\impliedby$ ) Let  $E$  be a compact set relative to  $X$ . Let  $\{H_\alpha\}_{\alpha \in A}$  be an open cover of  $E$  relative to  $Y$ . Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha.$$

By Theorem 2.4.23, there exists an open set  $G_\alpha$  in  $X$  such that  $H_\alpha = G_\alpha \cap Y$  for each  $\alpha \in A$ . Then we have

$$E \subseteq \bigcup_{\alpha \in A} H_\alpha = \bigcup_{\alpha \in A} (G_\alpha \cap Y) \subseteq \bigcup_{\alpha \in A} G_\alpha.$$

Thus  $\{G_\alpha\}_{\alpha \in A}$  is an open cover of  $E$  relative to  $X$ . Since  $E$  is compact relative to  $X$ , there exists a finite subcover  $\{G_\alpha\}_{\alpha \in S_A}$  of  $E$  relative to  $X$ . Then we have

$$E \subseteq \left( \bigcup_{\alpha \in S_A} G_\alpha \right) \cap Y = \bigcup_{\alpha \in S_A} (G_\alpha \cap Y) = \bigcup_{\alpha \in S_A} H_\alpha.$$

Thus  $\{H_\alpha\}_{\alpha \in S_A}$  is a finite subcover of  $E$  relative to  $Y$ . Therefore,  $E$  is compact relative to  $Y$ .  $\square$

*Remark.* The concept of **compactness** is preserved under the subspace topology, while the concepts of **openness** and **closedness** are not, which is the reason why

“Let  $(X, d)$  be a metric space and  $E \subseteq X$ .”

this kind of statement is always included in theorems and definitions about metric spaces. In this textbook, you can assume that

- $E$  is open.  $\implies E$  is open relative to  $X$ .
- $E$  is closed.  $\implies E$  is closed relative to  $X$ .
- $E$  is compact.  $\implies E$  is compact relative to  $X$ .

if there is no specific mention of the subspace topology.

**Theorem 2.5.2.** *Let  $(X, d)$  be a metric space and  $E \subseteq X$ . If  $E$  is compact, then  $E$  is closed.*

*Proof.* Let  $E$  be compact. We will prove that  $E^c$  is open, which implies that  $E$  is closed by [Corollary 2.4.10](#). Let  $p \in E^c$  be given. Since  $p \notin E$ , we have

$$r_q := \frac{d(p, q)}{2} > 0 \quad \text{for each } q \in E.$$

Since  $q \in N_{r_q}(q)$  for each  $q \in E$ , we have

$$E \subseteq \bigcup_{q \in E} N_{r_q}(q) \implies \{N_{r_q}(q)\}_{q \in E} \text{ is an open cover of } E.$$

Since  $E$  is compact, there exists a finite subcover  $\{N_{r_q}(q)\}_{q \in S}$  of  $E$ . Let  $T$  be a set defined by

$$T := \{r_q \mid q \in S\}.$$

Let  $f : S \rightarrow T$  be a function defined by

$$f(q) = r_q.$$

Then  $f$  is surjective. By [Theorem 2.2.4](#),  $T$  is finite. Then, by [Theorem 2.2.5](#),  $T$  has a minimum element, say  $\varepsilon = \min T > 0$ . For any  $x \in N_\varepsilon(p)$ , we have

$$d(x, p) < \varepsilon \leq r_q \quad \text{for each } q \in S.$$

Assume that there exists some  $q \in S$  such that  $x \in N_{r_q}(q)$ . Then we have

$$d(p, q) \leq d(p, x) + d(x, q) < r_q + r_q = 2r_q = d(p, q),$$

which is a contradiction. Then the following statements hold:

- $N_\varepsilon(p) \cap N_{r_q}(q) = \emptyset$  for each  $q \in S$ .
- $E \subseteq \bigcup_{q \in S} N_{r_q}(q)$ .

Therefore, we have  $N_\varepsilon(p) \subseteq E^c$ . Since  $p$  is arbitrary,  $E^c$  is open.  $\square$

**Theorem 2.5.3.** *Let  $(X, d)$  be a metric space and  $E \subseteq X$ . If  $E$  is compact, then  $E$  is bounded.*

*Proof.* Let  $E$  be compact. Let  $p \in E$  be given. Let  $\{G_n\}_{n \in \mathbb{N}}$  be a collection of open sets defined by

$$G_n = N_n(p) \quad \text{for each } n \in \mathbb{N}.$$

Then we have

$$n \leq m \implies G_n \subseteq G_m \quad \text{for each } n, m \in \mathbb{N}.$$

For any  $q \in X$ , by [Theorem 1.5.1](#), there exists  $n \in \mathbb{N}$  such that  $n > d(p, q)$ . Thus we have

$$E \subseteq X \subseteq \bigcup_{n=1}^{\infty} G_n \implies \{G_n\}_{n \in \mathbb{N}} \text{ is an open cover of } E.$$

Since  $E$  is compact, there exists a finite subcover  $\{G_n\}_{n \in S}$  of  $E$ . Since  $S \subseteq \mathbb{N}$  is finite, by [Theorem 2.2.5](#),  $S$  has a greatest element, say  $m = \max S$ . Then we have

$$E \subseteq \bigcup_{n \in S} G_n \subseteq G_m = N_m(p),$$

which implies that  $E$  is bounded.  $\square$

**Theorem 2.5.4.** *Let  $(X, d)$  be a metric space and  $E \subseteq X$ . If  $E$  is compact, then any closed subset  $F \subseteq E$  is compact.*

*Proof.* Let  $\{H_\alpha\}_{\alpha \in A}$  be an open cover of  $F$ . We have the following statements:

- $F$  is closed.  $\implies F^c$  is open.
- $F \subseteq \bigcup_{\alpha \in A} H_\alpha$ .

Then we have

$$X = F \cup F^c \subseteq \left( \bigcup_{\alpha \in A} H_\alpha \right) \cup F^c = \bigcup_{\alpha \in A} (H_\alpha \cup F^c).$$

$$\bullet E \subseteq X \implies E \subseteq \bigcup_{\alpha \in A} (H_\alpha \cup F^c).$$

By Theorem 2.4.11,  $H_\alpha \cup F^c$  is open for each  $\alpha \in A$ . Thus

$$\{H_\alpha \cup F^c\}_{\alpha \in A} \text{ is an open cover of } E.$$

Since  $E$  is compact, there exists a finite subcover  $\{H_\alpha \cup F^c\}_{\alpha \in B}$  of  $E$ . Then we have

$$\begin{aligned} \bullet E &\subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c). \\ \bullet F &\subseteq E \implies F \subseteq \bigcup_{\alpha \in B} (H_\alpha \cup F^c) = \left( \bigcup_{\alpha \in B} H_\alpha \right) \cup F^c. \end{aligned}$$

Since  $F \cap F^c = \emptyset$ , we have

$$F \subseteq \bigcup_{\alpha \in B} H_\alpha.$$

Thus  $\{H_\alpha\}_{\alpha \in B}$  is a finite subcover of  $F$ . Therefore,  $F$  is compact.  $\square$

**Theorem 2.5.5.** *Let  $(X, d)$  be a metric space and  $E \subseteq X$ . If  $E$  is compact, then any infinite subset of  $E$  has a limit point in  $E$ .*

*Proof.* Let  $E$  be compact. Let  $F \subseteq E$  be an infinite subset of  $E$ . Assume that  $F$  has no limit point in  $E$ . Then, for each  $p \in E$ , there exists  $r_p \in \mathbb{R}^+$  such that

$$\tilde{N}_{r_p}(p) \cap F = \emptyset \implies N_{r_p}(p) \cap F \subseteq \{p\}.$$

Let  $\{G_\alpha\}_{\alpha \in E}$  be a collection of open sets defined by

$$G_\alpha = N_{r_\alpha}(\alpha) \quad \text{for each } \alpha \in E.$$

Since  $\alpha \in G_\alpha$  for each  $\alpha \in E$ , we have

$$E \subseteq \bigcup_{\alpha \in E} G_\alpha \implies \{G_\alpha\}_{\alpha \in E} \text{ is an open cover of } E.$$

Since  $E$  is compact, there exists a finite subcover  $\{G_\alpha\}_{\alpha \in S}$  of  $E$ . Then we have

$$E \subseteq \bigcup_{\alpha \in S} G_\alpha \implies F \subseteq E \subseteq \bigcup_{\alpha \in S} G_\alpha.$$

Since  $F \subseteq \bigcup_{\alpha \in S} G_\alpha$ , we have

$$F = F \cap \bigcup_{\alpha \in S} G_\alpha = \bigcup_{\alpha \in S} (F \cap G_\alpha) \subseteq \bigcup_{\alpha \in S} \{\alpha\} = S,$$

which is a contradiction by Theorem 2.2.2. Therefore, any infinite subset of  $E$  has a limit point in  $E$ .  $\square$

**Theorem 2.5.6.** *Let  $(X, d)$  be a metric space and let  $\{K_\alpha\}_{\alpha \in A}$  be a collection of compact sets. Then we have*

$$\bigcap_{\alpha \in B} K_\alpha \neq \emptyset \text{ for all finite subset } B \subseteq A \implies \bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

*Proof.* Assume that  $\bigcap_{\alpha \in A} K_\alpha = \emptyset$ . Then we have

$$X = \left( \bigcap_{\alpha \in A} K_\alpha \right)^c = \bigcup_{\alpha \in A} (K_\alpha)^c.$$

Since each  $K_\alpha$  is compact,  $K_\alpha$  is closed by [Theorem 2.5.2](#). Then, by [Corollary 2.4.10](#), we have

$$\text{Each } (K_\alpha)^c \text{ is open} \implies \{(K_\alpha)^c\}_{\alpha \in A} \text{ is an open cover of } X.$$

Since  $K_\alpha \subseteq X \subseteq \bigcup_{\beta \in A} (K_\beta)^c$ , we have

$$\{(K_\beta)^c\}_{\beta \in A} \text{ is an open cover of } K_\alpha.$$

Since  $K_\alpha$  is compact, there exists a finite subcover  $\{(K_\beta)^c\}_{\beta \in B_\alpha}$  of  $K_\alpha$ . Then we have

$$\begin{aligned} K_\alpha &\subseteq \bigcup_{\beta \in B_\alpha} (K_\beta)^c = \left( \bigcap_{\beta \in B_\alpha} K_\beta \right)^c \\ &\implies K_\alpha \cap \left( \bigcap_{\beta \in B_\alpha} K_\beta \right) = \emptyset \\ &\implies \bigcap_{\beta \in B_\alpha \cup \{\alpha\}} K_\beta = \emptyset. \end{aligned}$$

Since  $\{\alpha\}$  and  $B_\alpha$  are finite,  $B_\alpha \cup \{\alpha\}$  is also finite by [Theorem 2.2.11](#), which contradicts the assumption. Therefore, we have

$$\bigcap_{\alpha \in A} K_\alpha \neq \emptyset.$$

□

**Lemma 2.5.7.** *Let  $(a_n)_{n=1}^\infty$  and  $(b_n)_{n=1}^\infty$  be sequences in  $\mathbb{R}$ , and let  $\{I_n\}_{n=1}^\infty$  be a sequence of intervals in  $\mathbb{R}$  such that*

- $I_n = [a_n, b_n]$  for all  $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$  for all  $n \in \mathbb{N}$

where  $a_n \leq b_n$  for all  $n \in \mathbb{N}$ . Then we have

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

*Proof.* Let  $E$  be a set defined by

$$E := \{a_n \mid n \in \mathbb{N}\}.$$

Since  $E$  is bounded above by  $b_1$ , there exists  $\alpha = \sup E$ . Then we have

$$a_m \leq \alpha \quad \text{for all } m \in \mathbb{N}.$$

For any  $m \in \mathbb{N}$ , we have  $b_m$  is an upper bound of  $E$  since

- $a_n \leq a_m \leq b_m$  for all  $n \in \mathbb{N}$  such that  $n \leq m$ .
- $a_n \leq b_n \leq b_m$  for all  $n \in \mathbb{N}$  such that  $n > m$ .

Therefore, we have

$$a_m \leq \alpha \leq b_m \implies \alpha \in I_m \text{ for all } m \in \mathbb{N}.$$

□

**Theorem 2.5.8.** Let  $k \in \mathbb{N}$  be given. Let  $\{a_{n,j}\}_{n=1}^{\infty}$  and  $\{b_{n,j}\}_{n=1}^{\infty}$  be sequences in  $\mathbb{R}$  for all  $j = 1, 2, \dots, k$ , and let  $\{I_n\}_{n=1}^{\infty}$  be a sequence of  $k$ -cells in  $\mathbb{R}^k$  such that

- $I_n = \{\mathbf{x} \in \mathbb{R}^k \mid a_{n,j} \leq x_j \leq b_{n,j} \text{ for all } j = 1, 2, \dots, k\}$  for all  $n \in \mathbb{N}$
- $I_{n+1} \subseteq I_n \iff a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$  for all  $j = 1, 2, \dots, k$  and  $n \in \mathbb{N}$

where  $a_{n,j} \leq b_{n,j}$  for all  $j = 1, 2, \dots, k$  and  $n \in \mathbb{N}$ . Then we have

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset.$$

*Proof.* Let  $\{I_{n,j}\}_{n=1}^{\infty}$  be a sequence of intervals in  $\mathbb{R}$  defined by

$$I_{n,j} = [a_{n,j}, b_{n,j}] \quad \text{for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}.$$

Then we have  $I_{n+1,j} \subseteq I_{n,j}$  for all  $j = 1, 2, \dots, k$  and  $n \in \mathbb{N}$ . By [Lemma 2.5.7](#), we have

$$\bigcap_{n=1}^{\infty} I_{n,j} \neq \emptyset \quad \text{for all } j = 1, 2, \dots, k.$$

Let  $\alpha_j \in \bigcap_{n=1}^{\infty} I_{n,j}$  be given for all  $j = 1, 2, \dots, k$ . Then we have

$$\alpha_j \in I_{n,j} \implies a_{n,j} \leq \alpha_j \leq b_{n,j} \quad \text{for all } j = 1, 2, \dots, k \text{ and } n \in \mathbb{N}.$$

Let  $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_k) \in \mathbb{R}^k$ . Then we have

$$\alpha \in I_n \quad \text{for all } n \in \mathbb{N} \implies \alpha \in \bigcap_{n=1}^{\infty} I_n.$$

□

**Theorem 2.5.9.** Let  $(\mathbb{R}^k, \|\cdot\|)$  be given for some  $k \in \mathbb{N}$ . Then any  $k$ -cell is compact.

*Proof.* Let  $I$  be a  $k$ -cell in  $\mathbb{R}^k$ , and let  $I_1 := I$ . Then we have

$$I_1 = \prod_{j=1}^k [a_{1,j}, b_{1,j}] \subseteq \mathbb{R}^k,$$

where  $-\infty < a_{1,j} \leq b_{1,j} < +\infty$  for all  $j = 1, 2, \dots, k$ . Let  $I_{1,j,m}$  be an interval in  $\mathbb{R}$  defined by

$$I_{1,j,m} := \begin{cases} \left[ a_{1,j}, \frac{a_{1,j} + b_{1,j}}{2} \right] & \text{if } m = 1 \\ \left[ \frac{a_{1,j} + b_{1,j}}{2}, b_{1,j} \right] & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \dots, k.$$

Let  $S_k$  be a set defined by

$$S_k := \{1, 2\}^k = \left\{ \mathbf{m} \in \mathbb{R}^k \mid m_j \in \{1, 2\} \text{ for all } j = 1, 2, \dots, k \right\}.$$

By [Theorem 2.2.13](#),  $S_k$  is finite. We have

$$I_1 = \bigcup_{\mathbf{m} \in S_k} \left( \prod_{j=1}^k I_{1,j,m_j} \right).$$

Let  $\{G_\alpha\}_{\alpha \in A}$  be an open cover of  $I_1$ . Assume that  $I_1$  has no finite subcover. Then, for each  $\mathbf{m} \in S_k$ , we have

$$\prod_{j=1}^k I_{1,j,m_j} \subseteq I_1 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that  $\{G_\alpha\}_{\alpha \in A}$  is an open cover of  $\prod_{j=1}^k I_{1,j,m_j}$ . Assume that for each  $\mathbf{m} \in S_k$ , we have

$$\prod_{j=1}^k I_{1,j,m_j} \text{ has a finite subcover } \{G_\alpha\}_{\alpha \in A_{\mathbf{m}}}.$$

Then we have

$$\prod_{j=1}^k I_{1,j,m_j} \subseteq \bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha \quad \text{for all } \mathbf{m} \in S_k,$$

where  $\bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha$  is open by [Theorem 2.4.11](#). Then we have

$$I_1 \subseteq \bigcup_{\mathbf{m} \in S_k} \left( \bigcup_{\alpha \in A_{\mathbf{m}}} G_\alpha \right) = \bigcup_{\alpha \in A} G_\alpha,$$

where  $A := \bigcup_{\mathbf{m} \in S_k} A_{\mathbf{m}}$  is finite by [Theorem 2.2.11](#). Thus  $I_1$  has a finite subcover, which is a contradiction. Therefore, there exists  $\mathbf{m}_1 \in S_k$  such that

$$\prod_{j=1}^k I_{1,j,(m_1)_j} \text{ has no finite subcover.}$$

Let  $I_2$  be a  $k$ -cell in  $\mathbb{R}^k$  defined by

$$I_2 := \prod_{j=1}^k [a_{2,j}, b_{2,j}] \subseteq \mathbb{R}^k,$$

where  $a_{2,j}, b_{2,j} \in \mathbb{R}$  is defined by

$$\begin{aligned} \bullet \quad a_{2,j} &:= \begin{cases} a_{1,j} & \text{if } (m_1)_j = 1 \\ \frac{a_{1,j} + b_{1,j}}{2} & \text{if } (m_1)_j = 2 \end{cases} \\ \bullet \quad b_{2,j} &:= \begin{cases} \frac{a_{1,j} + b_{1,j}}{2} & \text{if } (m_1)_j = 1 \\ b_{1,j} & \text{if } (m_1)_j = 2 \end{cases} \end{aligned}$$

for all  $j = 1, 2, \dots, k$ . Then, for each  $j = 1, 2, \dots, k$ , we have

$$[a_{2,j}, b_{2,j}] = I_{1,j,(m_1)_j} \subseteq [a_{1,j}, b_{1,j}],$$

which implies that

$$I_2 = \prod_{j=1}^k I_{1,j,(m_1)_j} \subseteq I_1.$$

Let  $I_{2,j,m}$  be an interval in  $\mathbb{R}$  defined by

$$I_{2,j,m} := \begin{cases} \left[ a_{2,j}, \frac{a_{2,j} + b_{2,j}}{2} \right] & \text{if } m = 1 \\ \left[ \frac{a_{2,j} + b_{2,j}}{2}, b_{2,j} \right] & \text{if } m = 2 \end{cases} \quad \text{for all } j = 1, 2, \dots, k.$$

Then we have

$$I_2 = \bigcup_{\mathbf{m} \in S_k} \left( \prod_{j=1}^k I_{2,j,m_j} \right).$$

We have shown that  $\{G_\alpha\}_{\alpha \in A}$  is an open cover of  $I_2$ , and  $I_2$  has no finite subcover. Then, for each  $\mathbf{m} \in S_k$ , we have

$$\prod_{j=1}^k I_{2,j,m_j} \subseteq I_2 \subseteq \bigcup_{\alpha \in A} G_\alpha,$$

which implies that  $\{G_\alpha\}_{\alpha \in A}$  is an open cover of  $\prod_{j=1}^k I_{2,j,m_j}$ . Therefore, there exists  $\mathbf{m}_2 \in S_k$  such that

$$\prod_{j=1}^k I_{2,j,(\mathbf{m}_2)_j} \text{ has no finite subcover.}$$

By repeating this process, we can construct the following sequences:

- The sequence  $(\mathbf{m}_n)_{n=1}^\infty$  in  $S_k$ .
- The sequences  $(a_{n,j})_{n=1}^\infty$  and  $(b_{n,j})_{n=1}^\infty$  in  $\mathbb{R}$  for all  $j = 1, 2, \dots, k$ .
- The sequence  $(I_{n,j,m})_{n=1}^\infty$  of intervals in  $\mathbb{R}$  for all  $j = 1, 2, \dots, k$  and  $m = 1, 2$ .
- The sequence  $(I_n)_{n=1}^\infty$  of  $k$ -cells in  $\mathbb{R}^k$ .

We have the following properties:

- $I_{n+1} \subseteq I_n$  for all  $n \in \mathbb{N}$ .
- $I_n$  has no finite subcover for all  $n \in \mathbb{N}$ .
- $a_{n,j} \leq a_{n+1,j} \leq b_{n+1,j} \leq b_{n,j}$  for all  $j = 1, 2, \dots, k$  and  $n \in \mathbb{N}$ .
- $b_{n+1,j} - a_{n+1,j} = \frac{1}{2}(b_{n,j} - a_{n,j}) \implies b_{n,j} - a_{n,j} = \frac{1}{2^{n-1}}(b_{1,j} - a_{1,j})$  for all  $j = 1, 2, \dots, k$  and  $n \in \mathbb{N}$ .

Let  $\delta := \sqrt{\sum_{j=1}^k (b_{1,j} - a_{1,j})^2}$ . Then we have

$$\|x - y\| \leq \sqrt{\sum_{j=1}^k (b_{n,j} - a_{n,j})^2} = \frac{\delta}{2^{n-1}} \quad \text{for all } x, y \in I_n \text{ and } n \in \mathbb{N}.$$

By [Theorem 2.5.8](#), we have  $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ . Let  $\beta \in \bigcap_{n=1}^{\infty} I_n$  be given. Then there exists  $\alpha_0 \in A$  such that  $\beta \in G_{\alpha_0}$ . Since  $G_{\alpha_0}$  is open, there exists  $\varepsilon \in \mathbb{R}^+$  such that

$$B_\varepsilon(\beta) \subseteq G_{\alpha_0}.$$

By [Corollary 1.5.2](#), there exists  $N \in \mathbb{N}$  such that  $\frac{1}{2^N} < \frac{\varepsilon}{2\delta}$ . Since  $\beta \in I_N$ , we have

$$\|\beta - \gamma\| \leq \frac{\delta}{2^{N-1}} < \varepsilon \quad \text{for all } \gamma \in I_N,$$

which implies that  $I_N \subseteq B_\varepsilon(\beta) \subseteq G_{\alpha_0}$ . Therefore,  $I_N$  has a finite subcover  $\{G_{\alpha_0}\}$ , which is a contradiction. Thus  $I$  has a finite subcover, and  $I$  is compact.  $\square$

**Corollary 2.5.10.** *For any  $-\infty < a \leq b < +\infty$ , the closed interval  $[a, b]$  is compact in  $(\mathbb{R}, \|\cdot\|)$ .*

**Theorem 2.5.11.** *Let  $(\mathbb{R}^n, \|\cdot\|)$  be given for some  $n \in \mathbb{N}$  and  $E \subseteq \mathbb{R}^n$ . Then we have*

$$E \text{ is compact} \iff E \text{ is closed and bounded.}$$

*This theorem is called the **Heine-Borel Theorem**.*

*Proof.* ( $\implies$ ) Let  $E$  be compact.  $E$  is closed by [Theorem 2.5.2](#), and  $E$  is bounded by [Theorem 2.5.3](#).

( $\impliedby$ ) Let  $E$  be closed and bounded. Then there exists some  $p \in \mathbb{R}^n$  and  $r \in \mathbb{R}^+$  such that  $E \subseteq B_r(p)$ . Let  $I$  be a  $n$ -cell in  $\mathbb{R}^n$  defined by

$$I := \prod_{j=1}^n [p_j - r, p_j + r] \subseteq \mathbb{R}^n.$$

Then we have  $E \subseteq B_r(p) \subseteq I$ . By [Theorem 2.5.9](#),  $I$  is compact. Since  $E$  is closed, by [Theorem 2.5.4](#),  $E$  is compact.  $\square$

## 2.6 Connected Sets

Let  $(X, d)$  be a metric space and  $E \subseteq X$ .  $E$  is called **connected** if there exists no pair of sets  $A, B \subseteq X$  such that

- $\emptyset \neq A \subseteq E$  and  $\emptyset \neq B \subseteq E$ .
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$ .
- $E = A \cup B$ .

Such pair of sets  $A$  and  $B$  is called **separated**. If  $E$  is not connected, then  $E$  is called **disconnected**.

**Theorem 2.6.1.** *Let  $(\mathbb{R}, \|\cdot\|)$  be given. Then  $E \subseteq \mathbb{R}$  is connected if and only if*

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

*Proof.* ( $\implies$ ) Let  $E \subseteq \mathbb{R}$  be connected. Assume that there exist  $a, b \in E$  and  $x \in \mathbb{R}$  such that

$$a < x < b \text{ and } x \notin E.$$

Let  $A, B$  be sets defined by

- $A := E \cap (-\infty, x) \subseteq E$ .
- $B := E \cap (x, \infty) \subseteq E$ .

Then we have

- $A \neq \emptyset$  and  $B \neq \emptyset$  since  $a \in A$  and  $b \in B$ .
- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$ .
- $E = A \cup B$  since  $x \notin E$ .

Thus  $A$  and  $B$  are separated, which is a contradiction. Therefore, we have

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

( $\impliedby$ ) Let  $E \subseteq \mathbb{R}$  be a set such that

$$a < x < b \implies x \in E \quad \text{for all } a, b \in E \text{ and } x \in \mathbb{R}.$$

Assume that  $E$  is disconnected. Then there exist separated sets  $A, B \subseteq \mathbb{R}$  such that

- $\emptyset \neq A \subseteq E$  and  $\emptyset \neq B \subseteq E$ .

- $\overline{A} \cap B = A \cap \overline{B} = \emptyset$ .
- $E = A \cup B$ .

Let  $a \in A$  and  $b \in B$  be given. Without loss of generality, assume that  $a < b$ . Let

$$\alpha := \sup(A \cap [a, b]) \in \mathbb{R},$$

where we have

- $a \in A \implies A \cap [a, b] \neq \emptyset$ .
- $A \cap [a, b]$  is bounded above by  $b$ .

Then the *existence* of  $\alpha$  is guaranteed by the *least upper bound property*. By [Theorem 2.4.17](#), we have

$$\alpha \in \overline{(A \cap [a, b])} = (A \cap [a, b]) \cup (A \cap [a, b])'.$$

By [Definition 2.4.4](#), it is clear that

$$(A \cap [a, b])' \subseteq A'.$$

Thus we have

$$\alpha \in (A \cap [a, b]) \subseteq A \quad \text{or} \quad \alpha \in (A \cap [a, b])' \subseteq A' \implies \alpha \in \overline{A}.$$

Then the following statements hold:

- $\alpha = \sup(A \cap [a, b]) \implies a \leq \alpha \leq b$ .
- $\alpha \in \overline{A} \implies \alpha \notin B \implies \alpha \neq b$ .

If  $\alpha \notin A$ , then we have

- $\alpha \neq a \implies a < \alpha < b$
- $\alpha \notin A$  and  $\alpha \notin B \implies \alpha \notin E$

which is a contradiction. Otherwise, we have

$$\alpha \in A \implies \alpha \notin \overline{B}.$$

Then there exists  $r \in (0, b - \alpha)$  such that

$$\tilde{N}_r(\alpha) \cap B = \emptyset \implies \beta := \alpha + r \notin B.$$

Then we have

- $a \leq \alpha < \beta < b$
- $\beta > \alpha \implies \beta \notin A \implies \beta \notin E$

which is a contradiction. Therefore, we conclude that  $E$  is connected.  $\square$

## Chapter 3

# Sequences and Series

### 3.1 Convergent Sequences

Let  $(X, d)$  be a metric space. A **sequence**  $(x_n)_{n=1}^{\infty}$  is a function  $f : \mathbb{N} \rightarrow X$ , where  $x_n = f(n)$  for each  $n \in \mathbb{N}$ .

A sequence  $(x_n)_{n=1}^{\infty}$  is said to be **convergent** if there exists  $x \in X$  such that

For any  $\varepsilon \in \mathbb{R}^+$ , there exists  $N \in \mathbb{N}$  such that  $n \geq N \implies d(x_n, x) < \varepsilon$ .

In this case, we say that

- $(x_n)_{n=1}^{\infty}$  **converges** to  $x$ .
- $x$  is the **limit** of  $(x_n)_{n=1}^{\infty}$ .

We denote the limit of  $(x_n)_{n=1}^{\infty}$  by

- $\lim_{n \rightarrow \infty} x_n = x$ .
- $x_n \rightarrow x$  as  $n \rightarrow \infty$ .

If  $(x_n)_{n=1}^{\infty}$  is not convergent, it is said to be **divergent**.

**Example 3.1.1.** Let  $(\mathbb{R}, \|\cdot\|)$  be given. Let  $(x_n)_{n=1}^{\infty}$  be a sequence defined by

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

Then, for each  $\varepsilon \in \mathbb{R}^+$ , we have

$$n > \frac{1}{\varepsilon} \implies \|x_n - 0\| = \left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \varepsilon.$$

Therefore,  $(x_n)_{n=1}^{\infty}$  converges to 0.

Let  $(\mathbb{R}^+, \|\cdot\|)$  be a subspace of  $(\mathbb{R}, \|\cdot\|)$ . Let  $(x_n)_{n=1}^\infty$  be a sequence defined by

$$x_n = \frac{1}{n} \quad \text{for each } n \in \mathbb{N}.$$

By [Theorem 1.5.1](#), for each  $y \in \mathbb{R}^+$ , there exists  $N_y \in \mathbb{N}$  such that  $N_y > \frac{1}{y}$ . Let

$$\varepsilon_y := y - \frac{1}{N_y} \in \mathbb{R}^+ \quad \text{for each } y \in \mathbb{R}^+.$$

Then we have

$$n > N_y \implies \|x_n - y\| = \left| \frac{1}{n} - y \right| > \left| \frac{1}{N_y} - y \right| = \varepsilon_y.$$

Therefore,  $(x_n)_{n=1}^\infty$  diverges in  $(\mathbb{R}^+, \|\cdot\|)$ .

*Remark.* Let  $(X, d)$  be a metric space. In this textbook,  $(x_n)_{n=1}^\infty$  represents a sequence in  $(X, d)$ , while  $\{x_n\}_{n=1}^\infty$  represents a set defined by

$$\{x_n\}_{n=1}^\infty := \{x_n \mid n \in \mathbb{N}\} \subseteq X.$$

**Definition 3.1.2.** Let  $(X, d)$  be a metric space. A sequence  $(x_n)_{n=1}^\infty$  is said to be **Bounded** if  $\{x_n\}_{n=1}^\infty$  is bounded.

**Theorem 3.1.3.** Let  $(X, d)$  be a metric space. Let  $(x_n)_{n=1}^\infty$  be a convergent sequence. Then we have

(A)  $(x_n)_{n=1}^\infty$  is bounded.

(B) The limit of  $(x_n)_{n=1}^\infty$  is unique.

*Proof.* Let  $x$  be the limit of  $(x_n)_{n=1}^\infty$ .

(A) Let  $\varepsilon = 1$ . Then there exists  $N \in \mathbb{N}$  such that

$$n \geq N \implies d(x_n, x) < 1.$$

Let  $S$  be a set defined by

$$S := \{d(x_n, x) \mid 1 \leq n \leq N\} \subseteq [0, \infty).$$

Let  $f : J_N \rightarrow S$  be a function defined by

$$f(k) = d(x_k, x) \quad \text{for all } k \in J_N.$$

Then  $f$  is surjective. By [Lemma 2.2.3](#),  $S$  is finite. Then, by [Theorem 2.2.5](#),  $S$  has a greatest element, say  $M = \max S$ . Then we have

$$d(x_n, x) < M + 1 \quad \text{for all } n \in \mathbb{N}.$$

Therefore,  $(x_n)_{n=1}^\infty$  is bounded.

(B) Let  $y$  be the limit of  $(x_n)_{n=1}^{\infty}$ . Assume that  $x \neq y$ . Then we have  $r := d(x, y) > 0$ . Let  $\varepsilon = \frac{r}{2}$ . Then we have

- There exists  $N_x \in \mathbb{N}$  such that  $n \geq N_x \implies d(x_n, x) < \frac{r}{2}$ .
- There exists  $N_y \in \mathbb{N}$  such that  $n \geq N_y \implies d(x_n, y) < \frac{r}{2}$ .

Thus we have

$$n \geq \max\{N_x, N_y\} \implies d(x, y) \leq d(x, x_n) + d(x_n, y) < r,$$

which is a contradiction. Therefore, we have  $x = y$ .

□

**Theorem 3.1.4.** Let  $(X, d)$  be a metric space. Then the following statements hold:

(A) A sequence  $(x_n)_{n=1}^{\infty}$  converges to  $x \in X$  if and only if

For any  $\varepsilon \in \mathbb{R}^+$ ,  $N_{\varepsilon}(x)$  contains all but finitely many terms of  $(x_n)_{n=1}^{\infty}$ .

(B) Let  $E \subseteq X$  be given. If  $x$  is a limit point of  $E$ , then there exists a sequence  $(x_n)_{n=1}^{\infty}$  in  $(X, d)$  such that

$$\lim_{n \rightarrow \infty} x_n = x.$$

*Proof.* (A) ( $\implies$ ) Let  $(x_n)_{n=1}^{\infty}$  converge to  $x \in X$ . Then, for each  $\varepsilon \in \mathbb{R}^+$ , there exists  $M_{\varepsilon} \in \mathbb{N}$  such that

$$n \geq M_{\varepsilon} \implies d(x_n, x) < \varepsilon,$$

which implies that

$$\{x_n\}_{n=M_{\varepsilon}}^{\infty} \subseteq N_{\varepsilon}(x).$$

( $\impliedby$ ) Let  $(x_n)_{n=1}^{\infty}$  be a sequence in  $(X, d)$  such that

For any  $\varepsilon \in \mathbb{R}^+$ ,  $N_{\varepsilon}(x)$  contains all but finitely many terms of  $(x_n)_{n=1}^{\infty}$ .

(B) Let  $E \subseteq X$  be given. Assume that  $x$  is a limit point of  $E$ . Then for any  $\varepsilon \in \mathbb{R}^+$ , we have  $(N_{\varepsilon}(x) - \{x\}) \cap E \neq \emptyset$ . Thus we can choose a sequence  $(y_n)_{n=1}^{\infty}$  in  $(X, d)$  such that

$$y_n \in (N_{\frac{1}{n}}(x) \setminus \{x\}) \cap E.$$

Then we have

$$d(y_n, x) < \frac{1}{n} \quad \text{for each } n \in \mathbb{N},$$

which implies that  $\lim_{n \rightarrow \infty} y_n = x$ .

□

- 3.2 Subsequences**
- 3.3 Cauchy Sequences**
- 3.4 Upper and Lower Limits**
- 3.5 Some Special Sequences**
- 3.6 Series**
- 3.7 Series of Nonnegative Terms**
- 3.8 The Number  $e$**
- 3.9 The Root and Ratio Tests**
- 3.10 Power Series**
- 3.11 Summation by Parts**
- 3.12 Absolute Convergence**
- 3.13 Addition and Multiplication of Series**
- 3.14 Rearrangements**

## **Chapter 4**

# **Continuity**

### **4.1 Limits of Functions**

### **4.2 Continuous Functions**

### **4.3 Continuity and Compactness**

### **4.4 Continuity and Connectedness**

### **4.5 Discontinuities**

### **4.6 Monotonic Functions**

### **4.7 Infinite Limits and Limits at Infinity**

## **Chapter 5**

# **Differentiation**

### **5.1 The Derivative of a Real Function**

### **5.2 Mean Value Theorems**

### **5.3 The Continuity of Derivatives**

### **5.4 L'Hospital's Rule**

### **5.5 Derivatives of Higher Order**

### **5.6 Taylor's Theorem**

### **5.7 Differentiation of Vector-valued Functions**

## **Chapter 6**

# **The Riemann-Stieltjes Integral**

### **6.1 Definition and Existence of the Integral**

### **6.2 Properties of the Integral**

### **6.3 Integration and Differentiation**

### **6.4 Integration of Vector-valued Functions**

### **6.5 Rectifiable Curves**

## **Chapter 7**

# **Sequences and Series of Functions**

**7.1 Discussion of Main Problem**

**7.2 Uniform Convergence**

**7.3 Uniform Convergence and Continuity**

**7.4 Uniform Convergence and Integration**

**7.5 Uniform Convergence and Differentiation**

**7.6 Equicontinuous Families of Functions**

**7.7 Equicontinuous Families of Functions**

**7.8 The Stone-Weierstrass Theorem**

## **Chapter 8**

# **Some Special Functions**

### **8.1 Power Series**

### **8.2 The Exponential and Logarithmic Functions**

### **8.3 The Trigonometric Functions**

### **8.4 The Algebraic Completeness of the Complex Field**

### **8.5 Fourier Series**

### **8.6 The Gamma Function**

## **Chapter 9**

# **Functions of Several Variables**

**9.1 Linear Transformations**

**9.2 Differentiation**

**9.3 The Contraction Principle**

**9.4 The Inverse Function Theorem**

**9.5 The Implicit Function Theorem**

**9.6 The Rank Theorem**

**9.7 Determinants**

**9.8 Derivatives of Higher Order**

**9.9 Differentiation of Integrals**

## **Chapter 10**

# **Integration of Differential Forms**

### **10.1 Integration**

### **10.2 Primitive Mappings**

### **10.3 Partitions of Unity**

### **10.4 Change of Variables**

### **10.5 Differential Forms**

### **10.6 Simplexes and Chains**

### **10.7 Stokes' Theorem**

### **10.8 Closed Forms and Exact Forms**

### **10.9 Vector Analysis**

## **Chapter 11**

# **The Lebesgue Theory**

### **11.1 Set Functions**

### **11.2 Construction of the Lebesgue Measure**

### **11.3 Measure Spaces**

### **11.4 Measurable Functions**

### **11.5 Simple Functions**

### **11.6 Integration**

### **11.7 Comparison with the Riemann Integral**

### **11.8 Integration of Complex Functions**

### **11.9 Functions of Class $\mathcal{L}^2$**

## **Appendix: Construction of $\mathbb{R}$**

# Bibliography

- [1] Walter Rudin, *Principles of Mathematical Analysis*, McGraw-Hill International Editions.
- [2] Kenneth Ross, *Elementary Analysis*, Springer Nature.
- [3] Steven Douglass, *Introduction to Mathematical Analysis*, Pearson.
- [4] Terence Tao, *Analysis I*, Springer Nature.